



Enclustra User Schematics

Mercury XU5 Revision 1.2

Disclaimer

User schematics do not include proprietary Enclustra design elements, such as power supply circuits. These circuits use optimized designs which are extremely compact and remain proprietary to Enclustra. The user manual provides all necessary information to use the module and its interfaces.

User schematics are provided after purchase of Enclustra hardware, and may also be provided in certain cases before purchase, to assist in product evaluation.
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Full schematics, including a full bill-of-materials, may be available through the purchase of a hardware licence for the product in question.
Please contact Enclustra Sales for more information.

Note: DNE = Do Not Equip (parts not equipped by default)

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Mercury XU5 SoC Module

Revision 1.2

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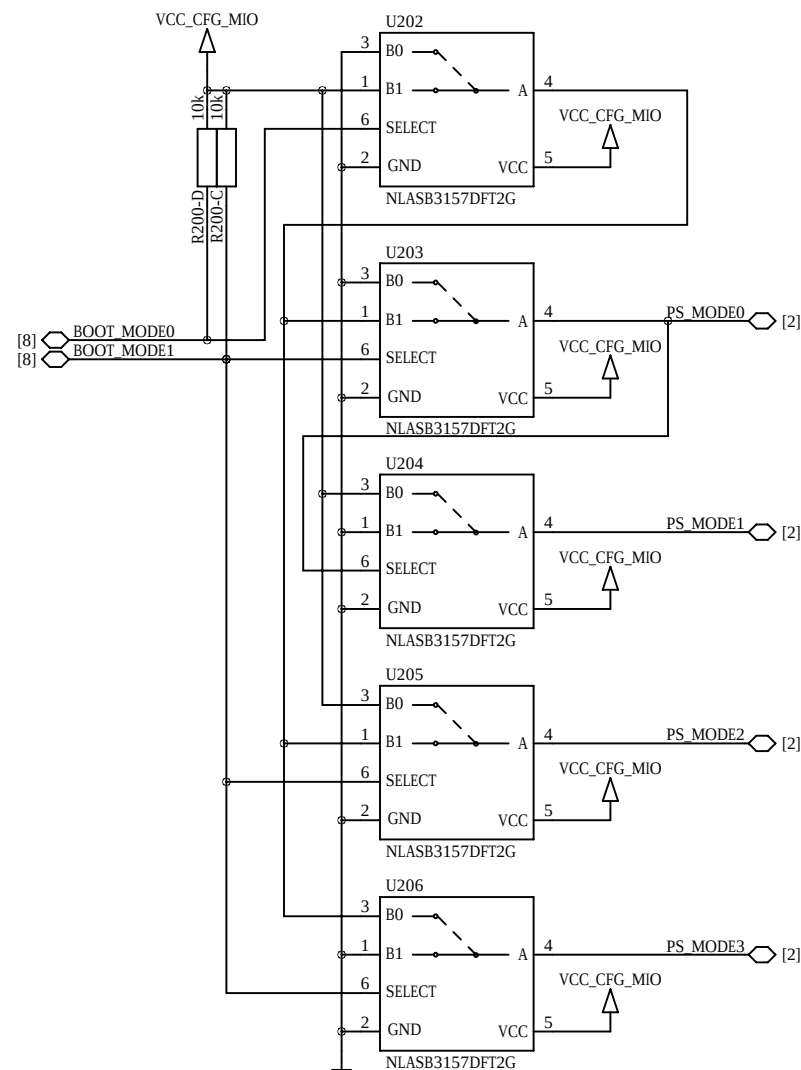
B

B

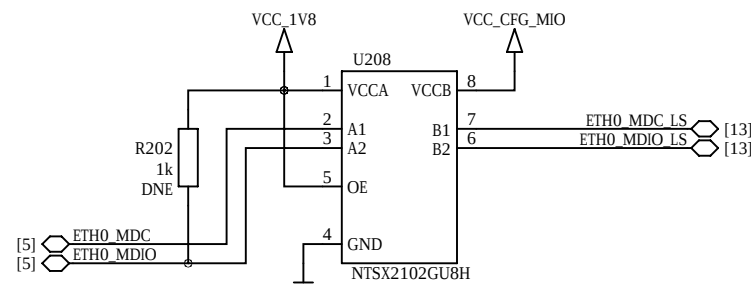
A

A

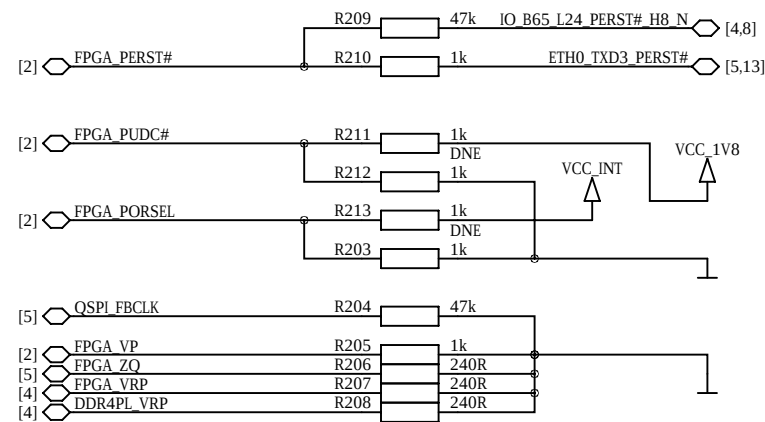
Boot Mode Selection



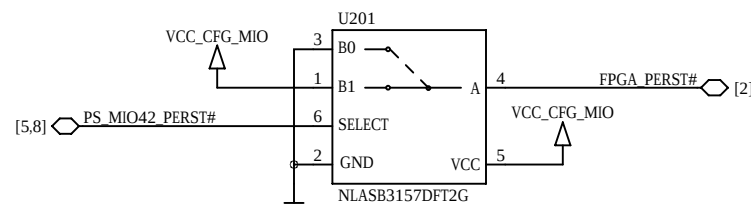
MDIO Level Shifter



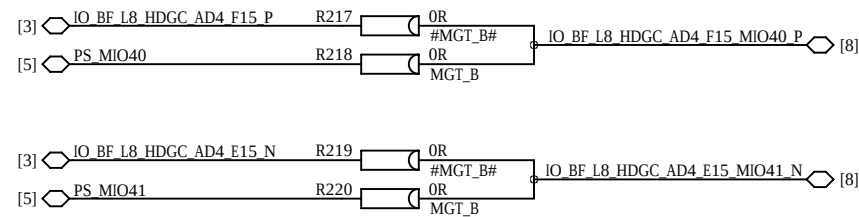
Miscellaneous



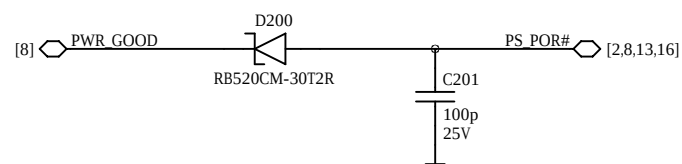
PCIe Reset



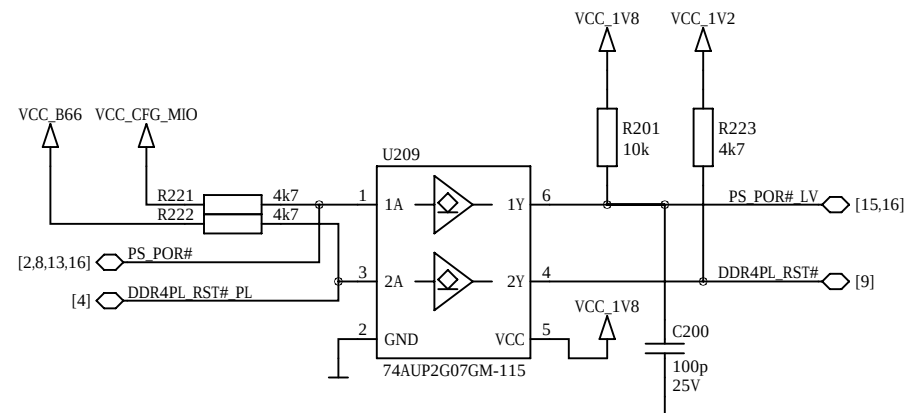
For compatibility with ZX1 / ZX5



Reset Circuit



Reset Level Shifter



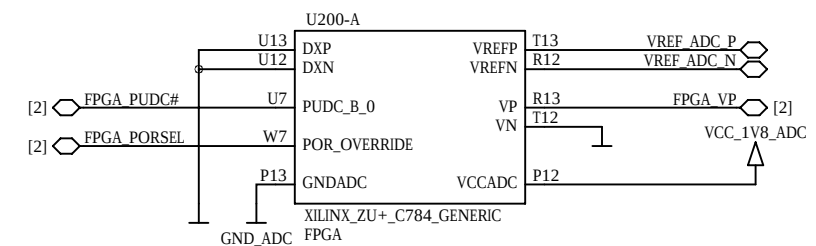
Boot Modes

	Boot Mode		Mode Straps		Configuration	
	1	0	PS_MODE	PL Boot	PS Boot	
eMMC Mode	0	0	0110	PS	eMMC	
SD Mode LS not supported	0	1	1110	PS	SDIO with level shifter	
QSPI Mode	1	0	0010	PS	QSPI	
SD Mode default	1	1	0101	PS	SDIO	

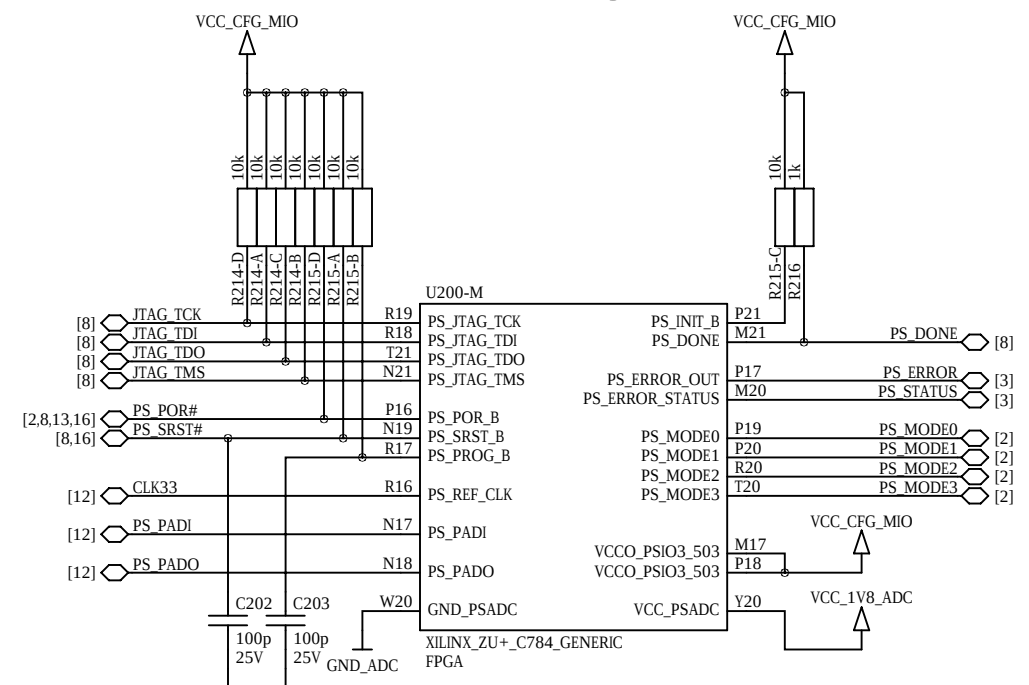
SD Mode LS mode may be supported in the future.

SD Mode LS requires VCC_CFG_MIO of 1.8V.

FPGA Configuration



FPGA PS Configuration

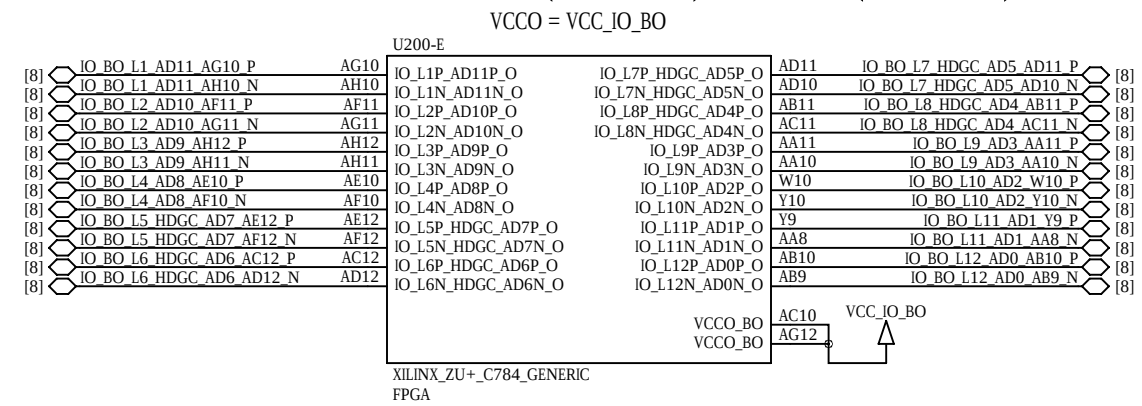
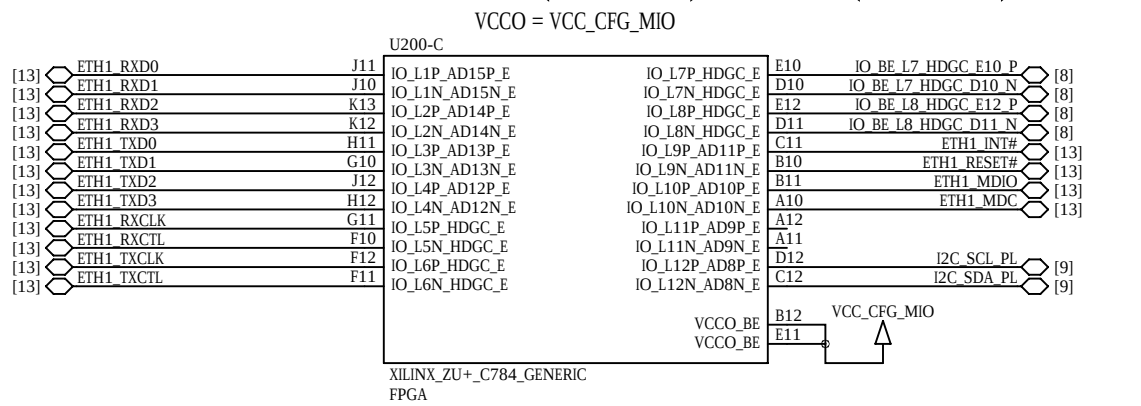


D

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FPGA HD Bank 25 (ZU2/3) or 45 (ZU4/5)

FPGA HD Bank 44 (ZU2/3) or 43 (ZU4/5)

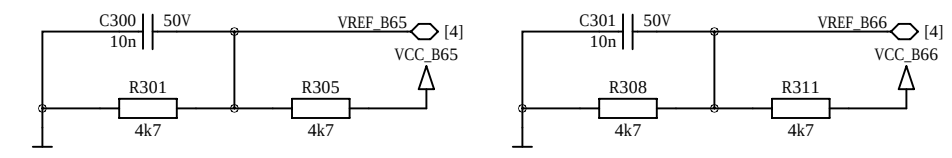
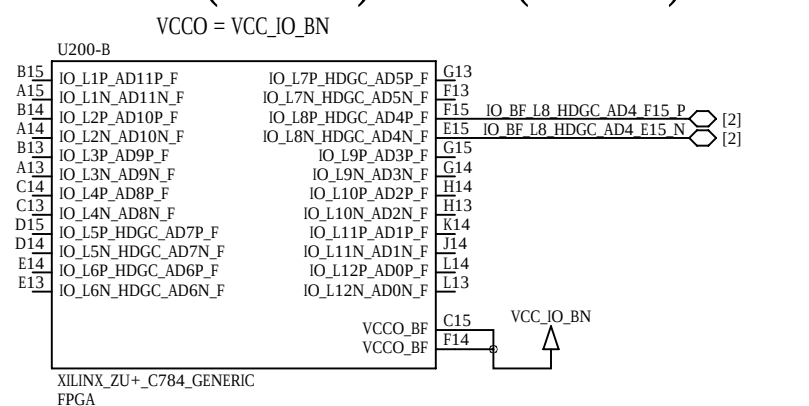


C

C

FPGA HD Bank 26 (ZU2/3) or 46 (ZU4/5)

FPGA Reference Voltages

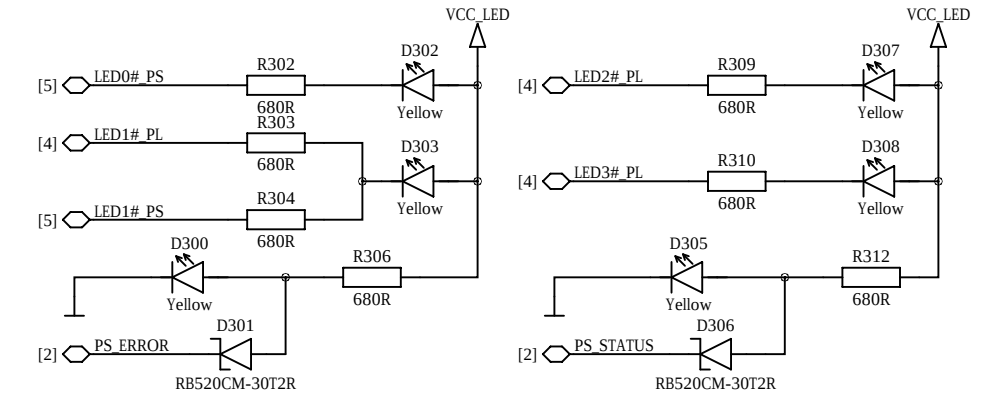
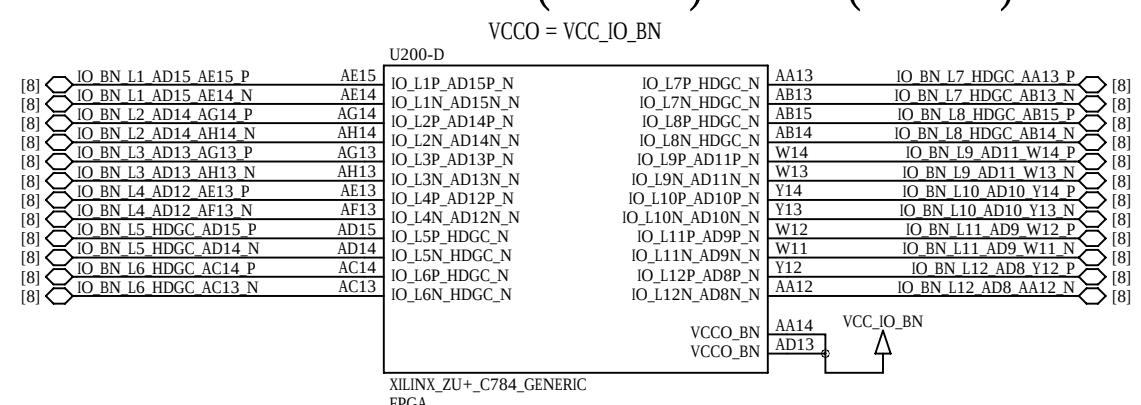


B

B

FPGA HD Bank 24 (ZU2/3) or 44 (ZU4/5)

User LEDs

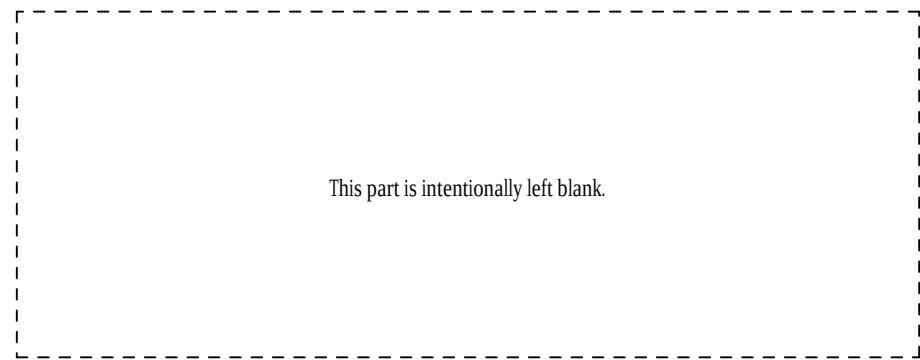


A

A

LED Power

not included in user schematics



6

5

4

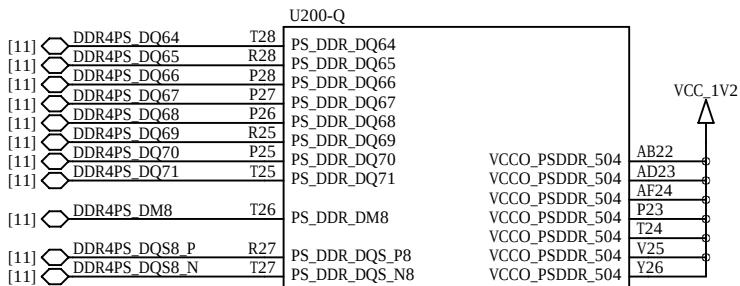
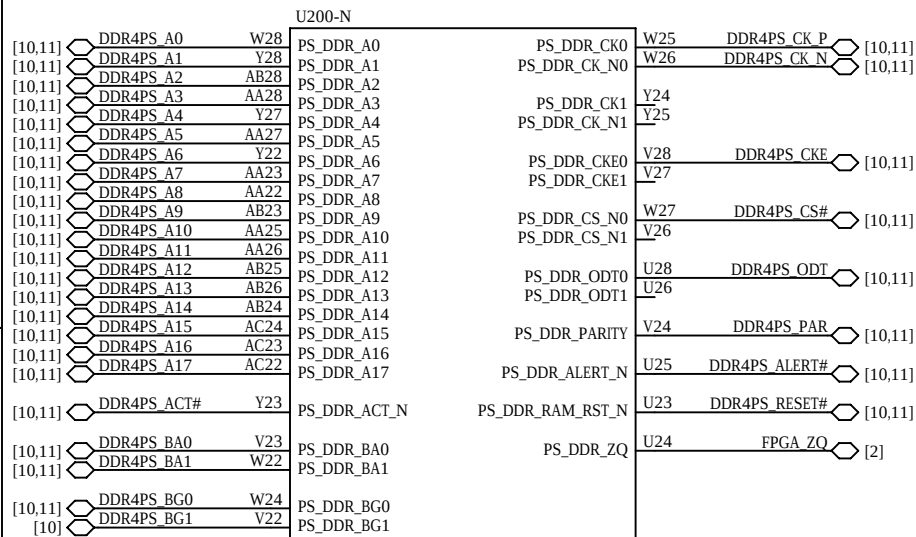
3

2

1

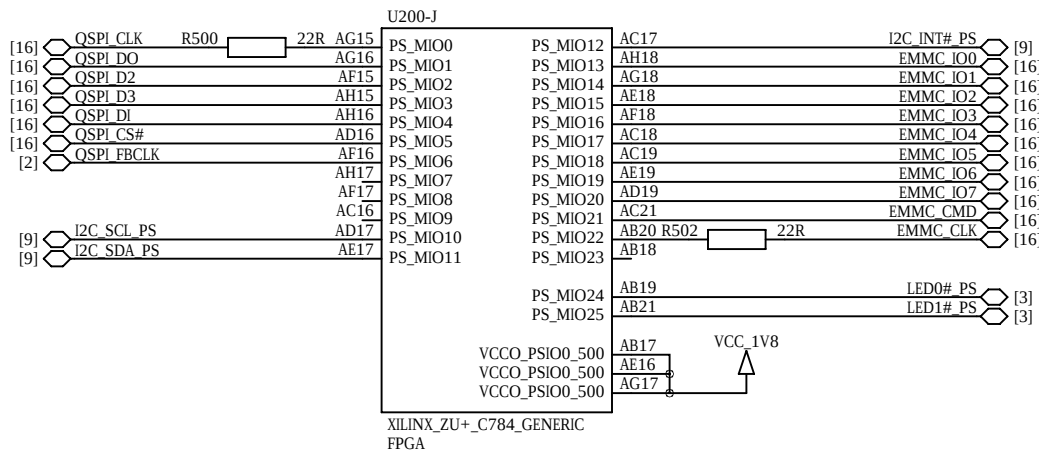
FPGA PS DDR Bank 504

VCCO = VCC_1V2

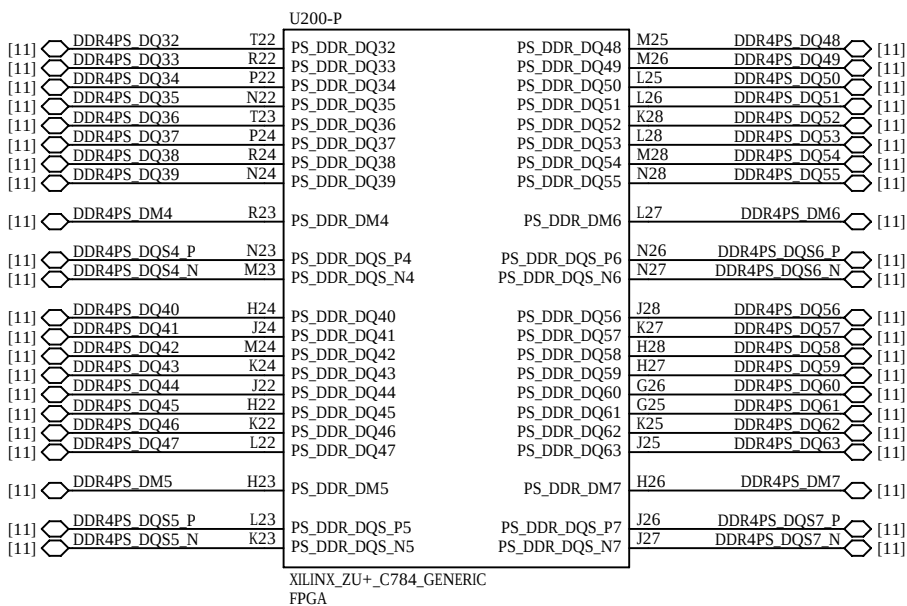
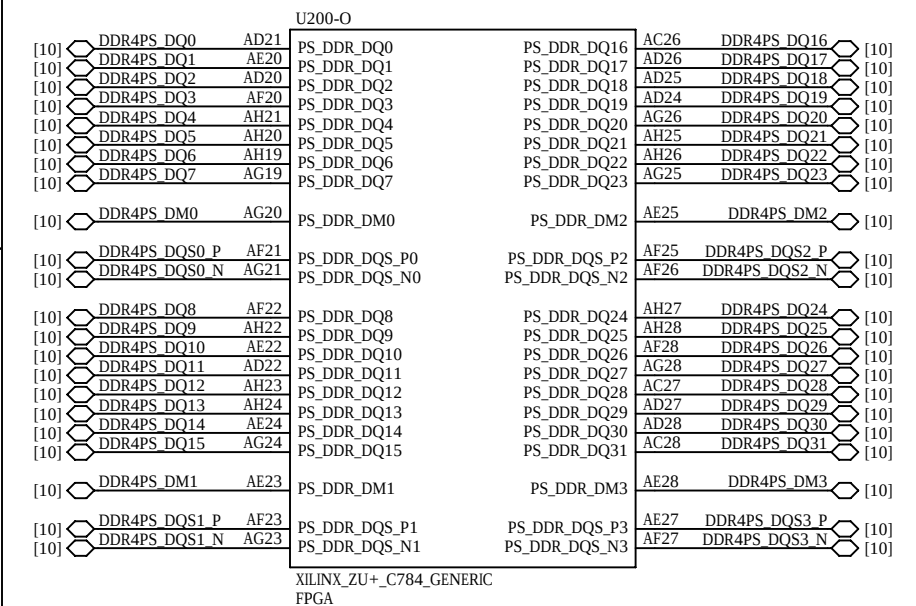


FPGA PS Bank 500

VCCO = VCC_1V8

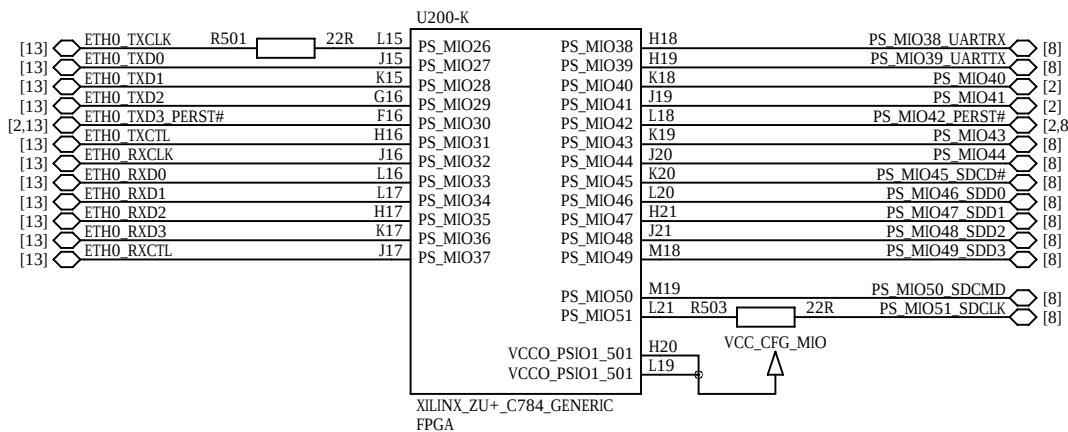


C



FPGA PS Bank 501

VCCO = VCC_CFG_MIO



B

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B

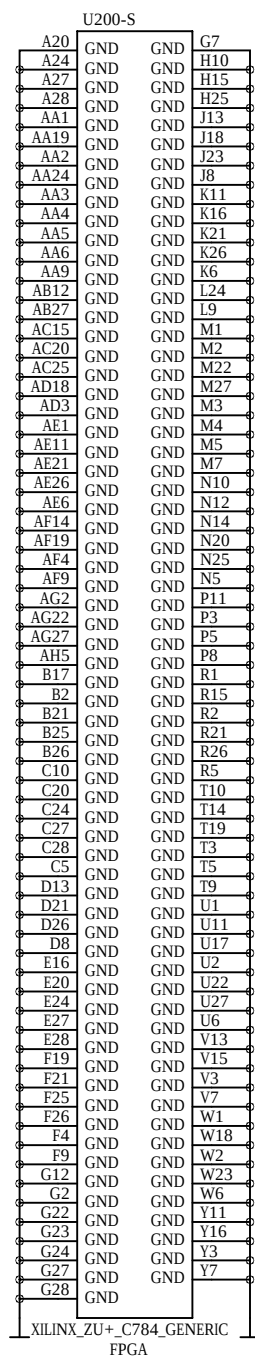
A

A

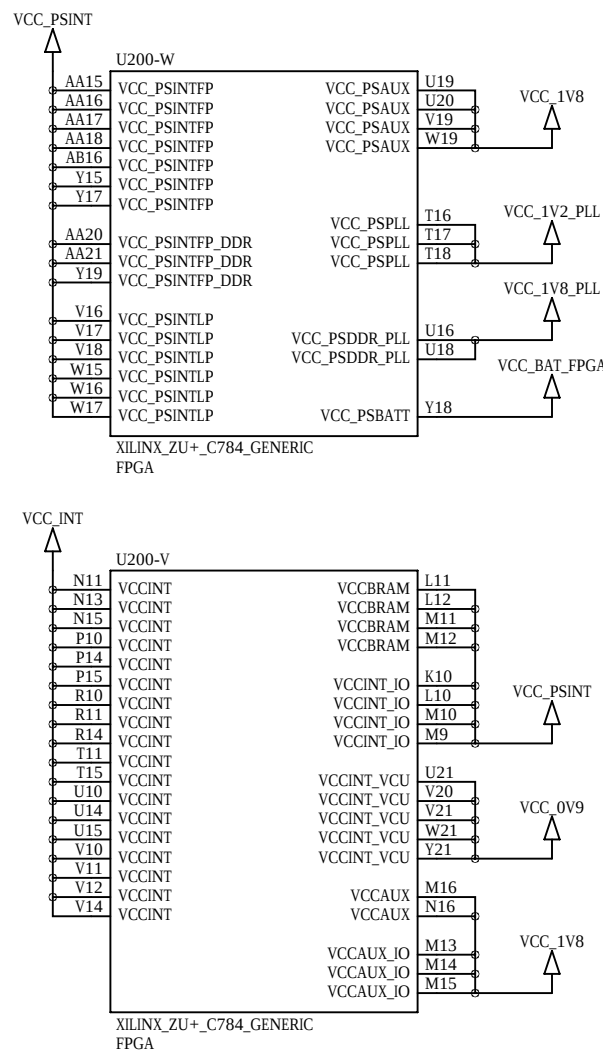


Copyright	© 2021 by Enclustra GmbH	Sheet Name	05_FPGA_PS_BANKS	Customer No	0000	Revision	R1.2	DNE = do not equip	Confidential		
Company	Enclustra FPGA Solution Center	Project	Mercury XU5	Project No	445	Designed	MHEI	Date	17 Mai 2021	Sheet/sheets	5 / 23

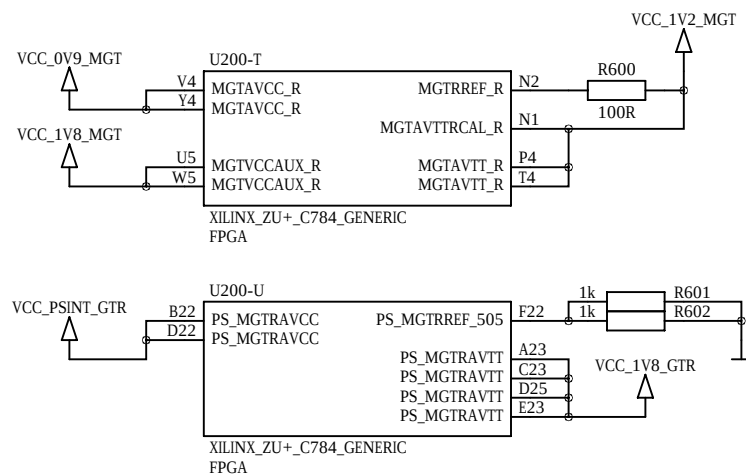
FPGA Ground



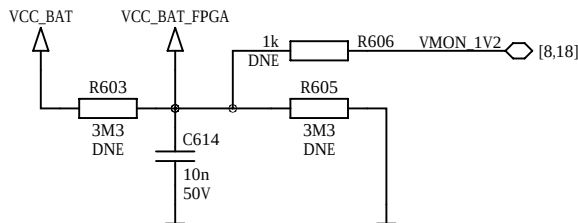
FPGA Power



FPGA MGT Power



FPGA Battery Voltage



ADC Power

not included in user schematics

This part is intentionally left blank.

ADC Reference

optional

not included in user schematics

This part is intentionally left blank.

MGT Power Filters

not included in user schematics

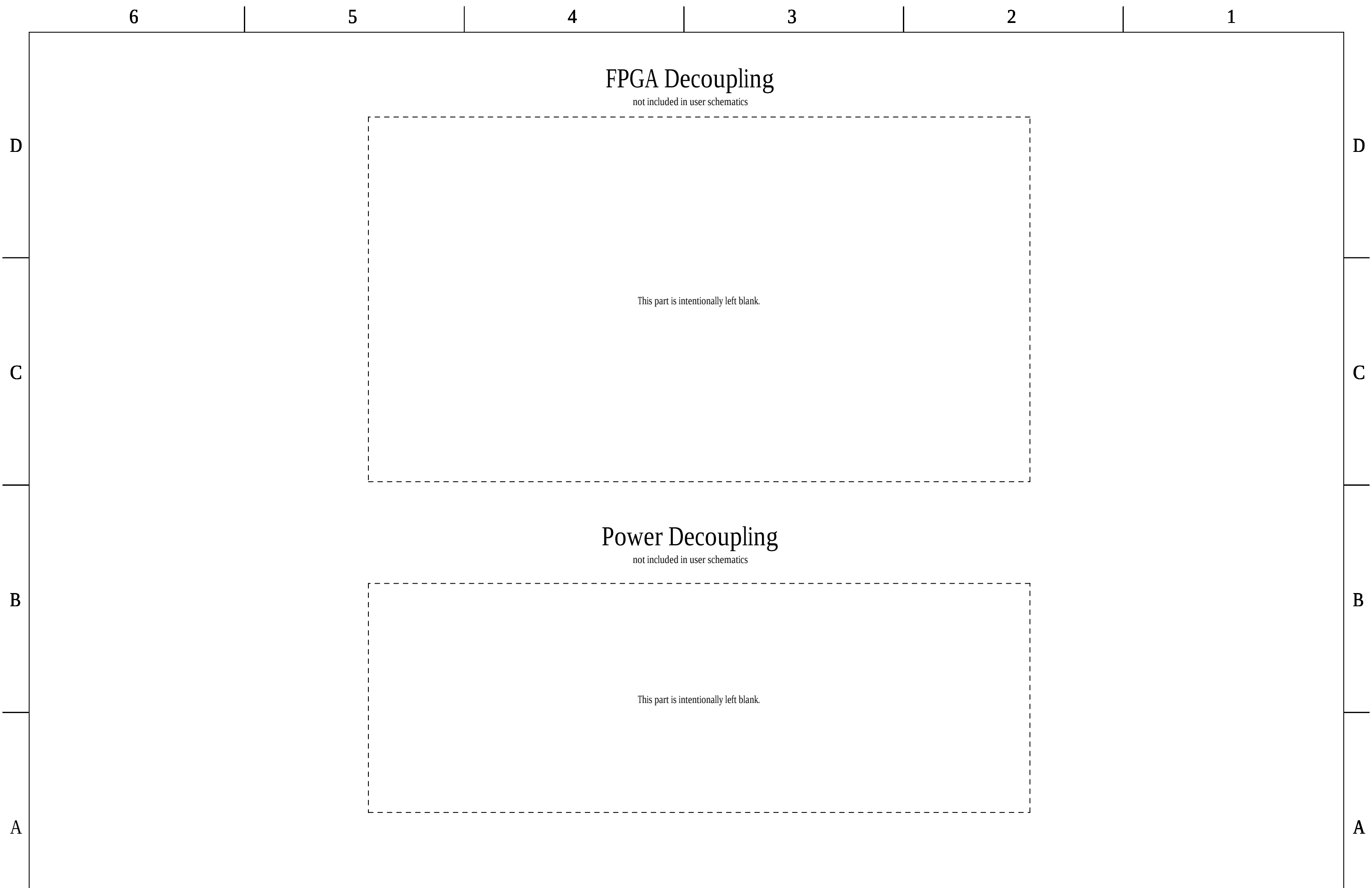
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PLL and GTR Power Filters

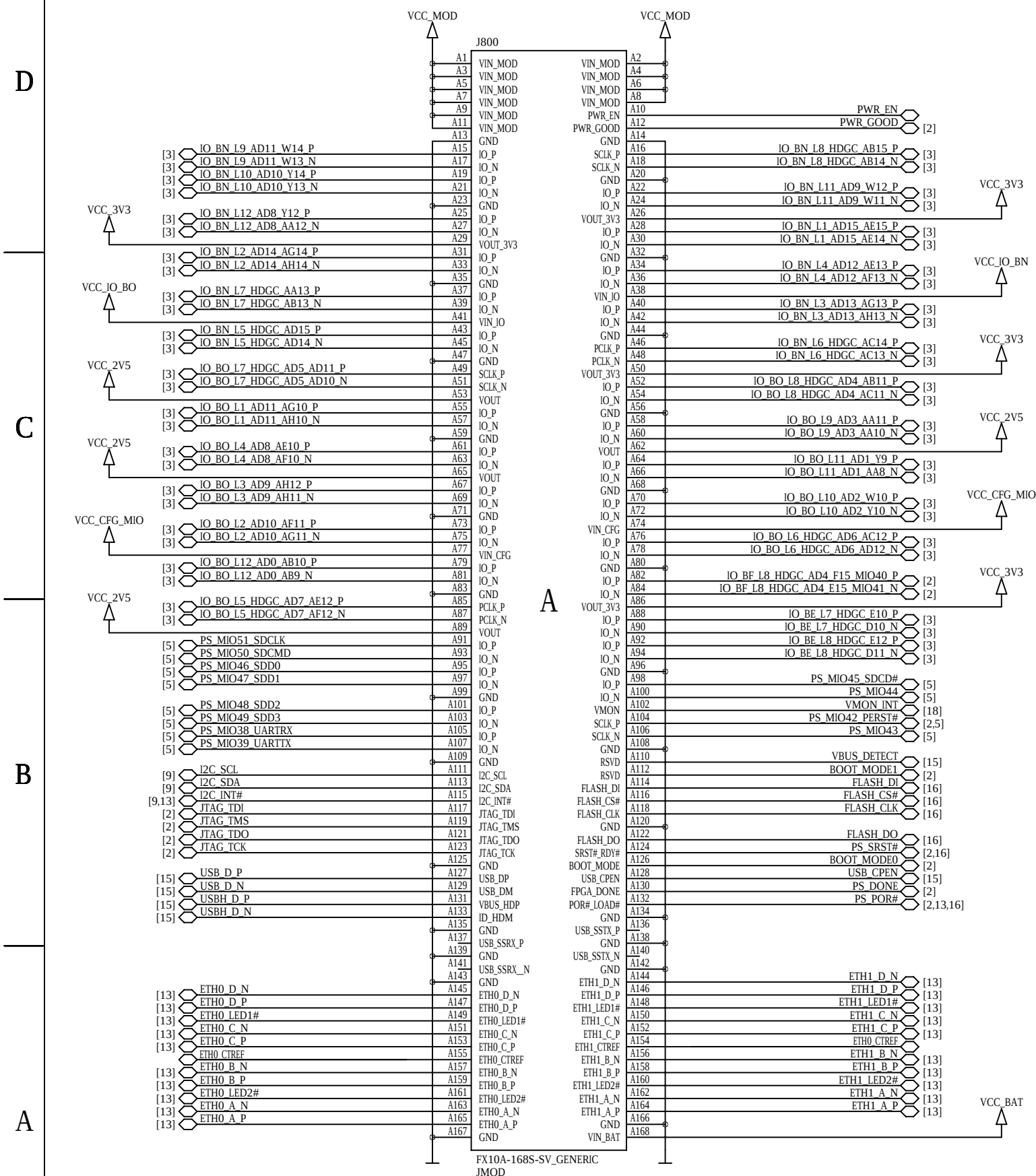
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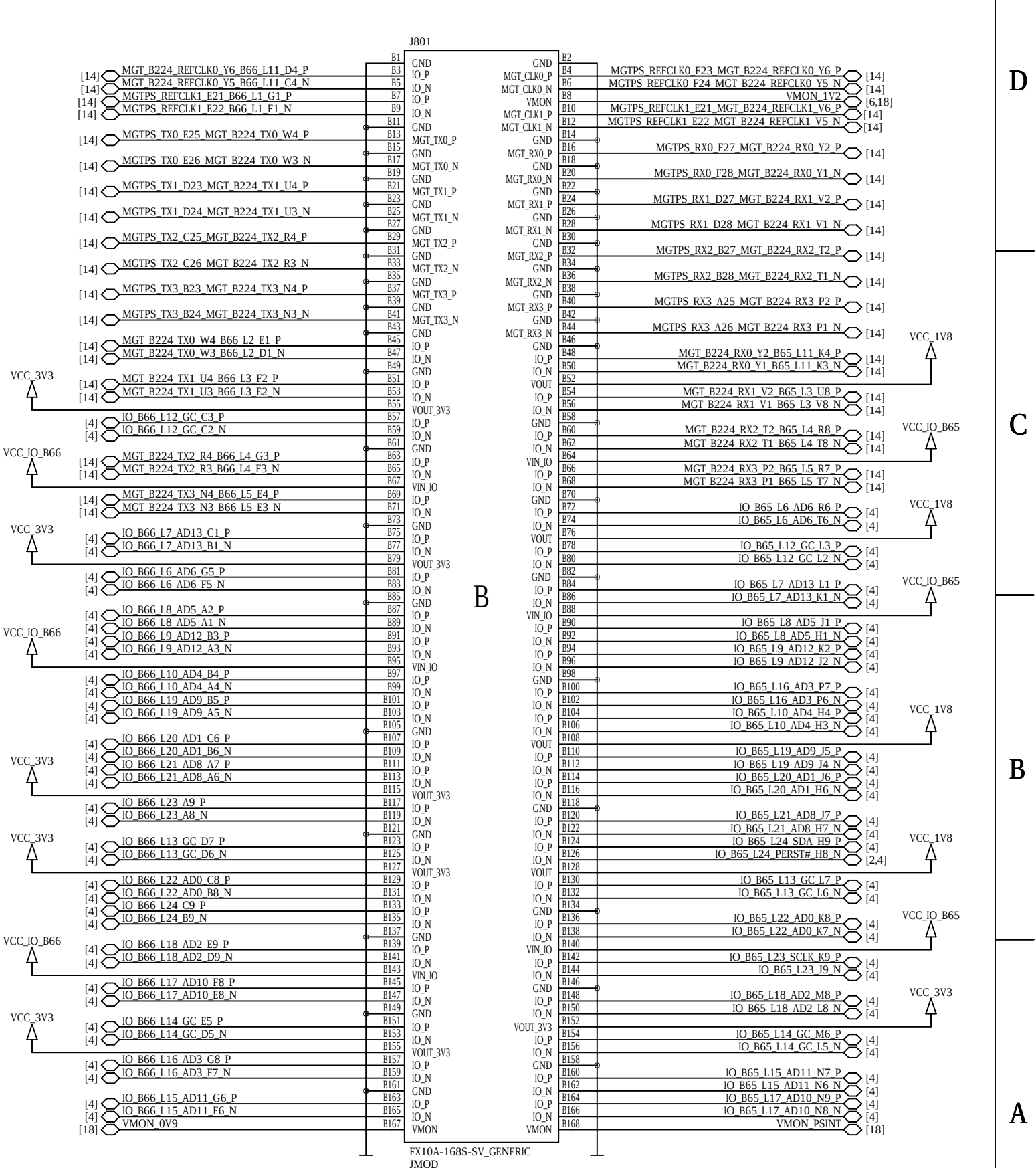




Module Connector A

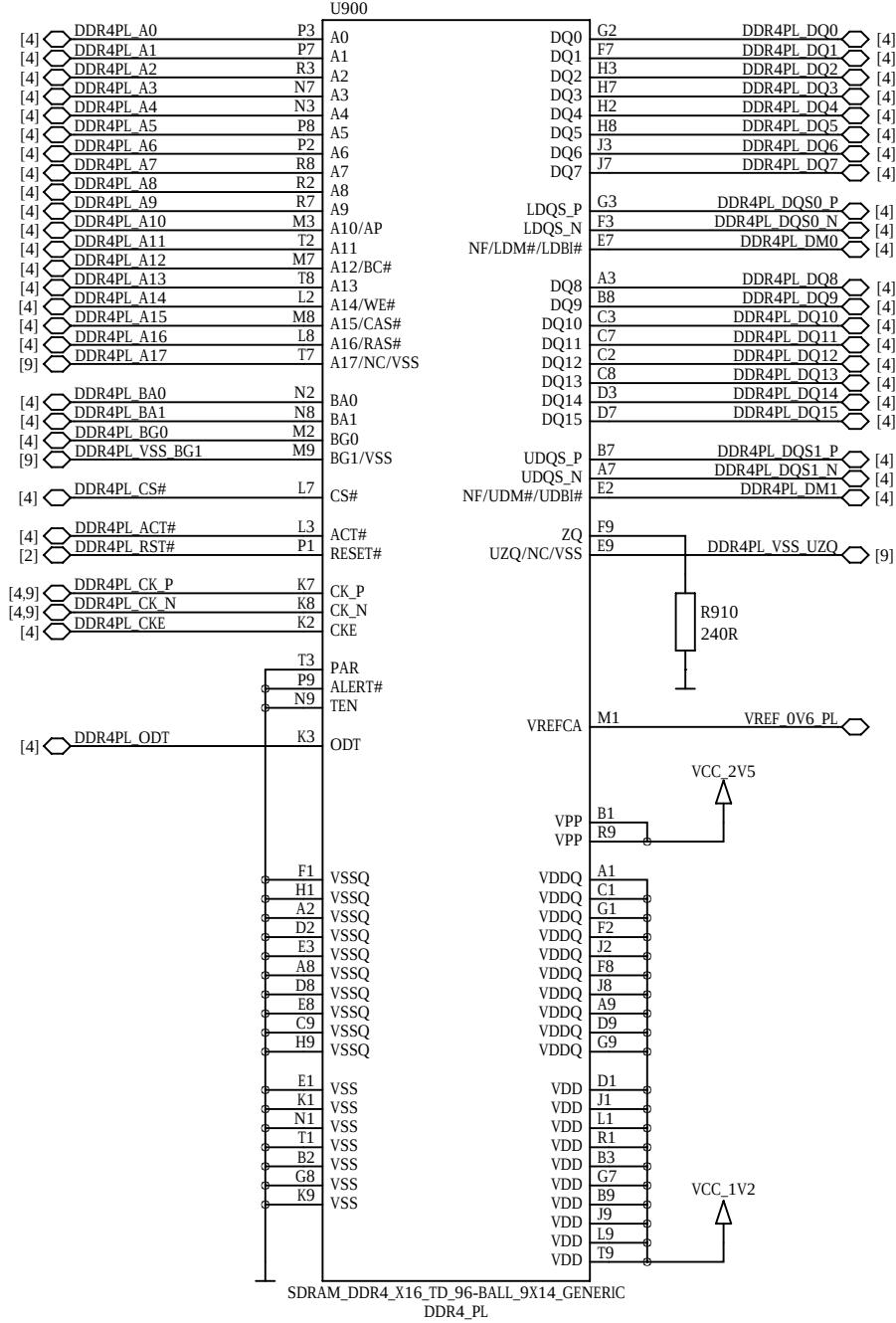


Module Connector B

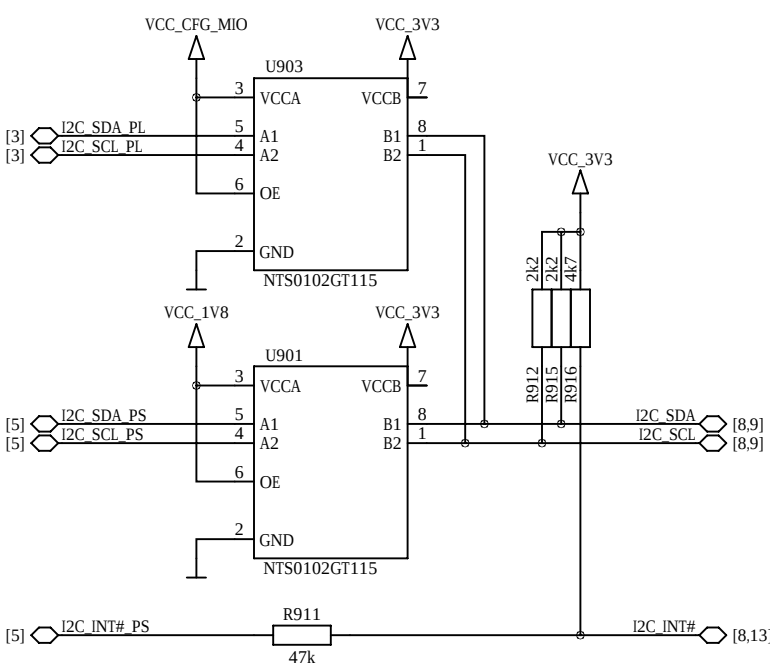


Copyright	© 2021 by Enclustra GmbH	Sheet Name	08_MODULE_CONNECTORS_A_B	Customer No	0000	Revision	R1.2	DNE = do not equip	Confidential		
Company	Enclustra FPGA Solution Center	Project	Mercury XU5	Project No	445	Designed	MHEI	Date	17 Mai 2021	Sheet/sheets	8 / 23

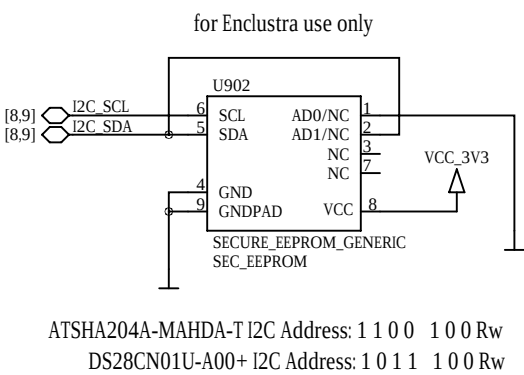
PL DDR4 SDRAM



I2C Level Shifter



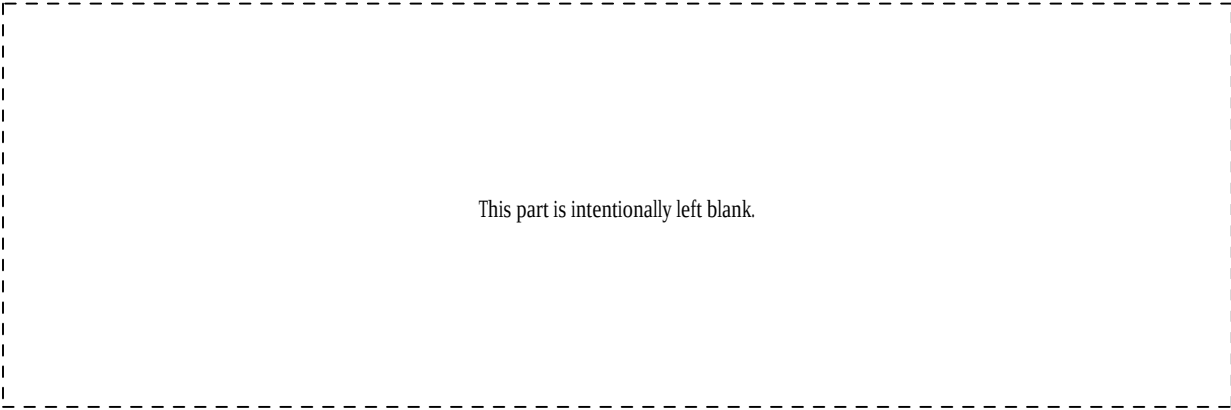
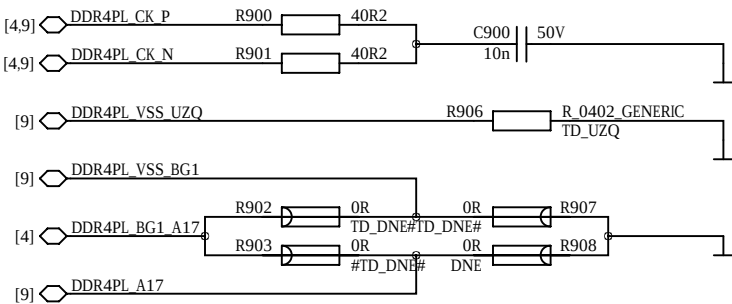
Secure EEPROM



PL DDR4 VREF & Decoupling

not included in user schematics

PL DDR4 Miscellaneous



D

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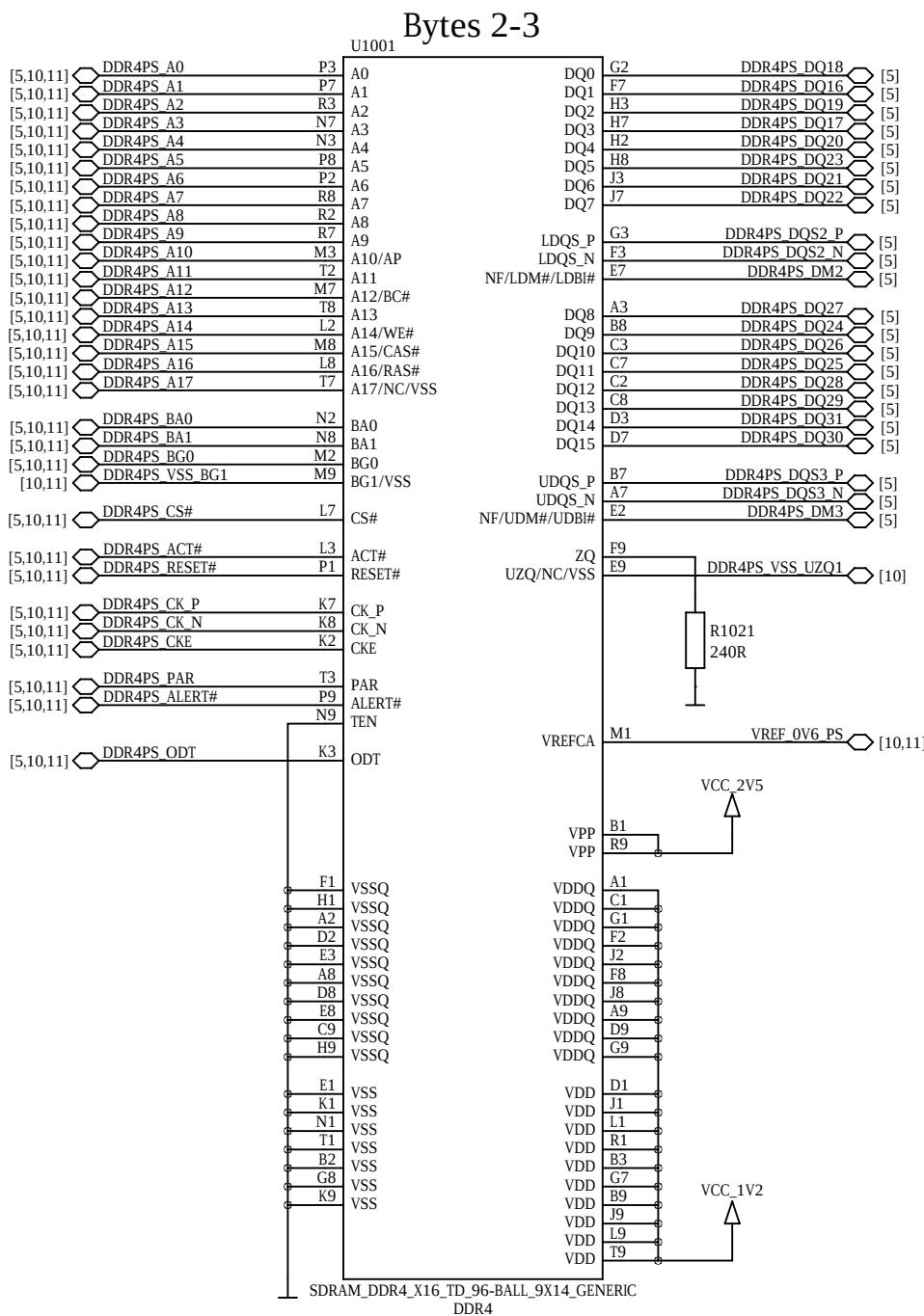
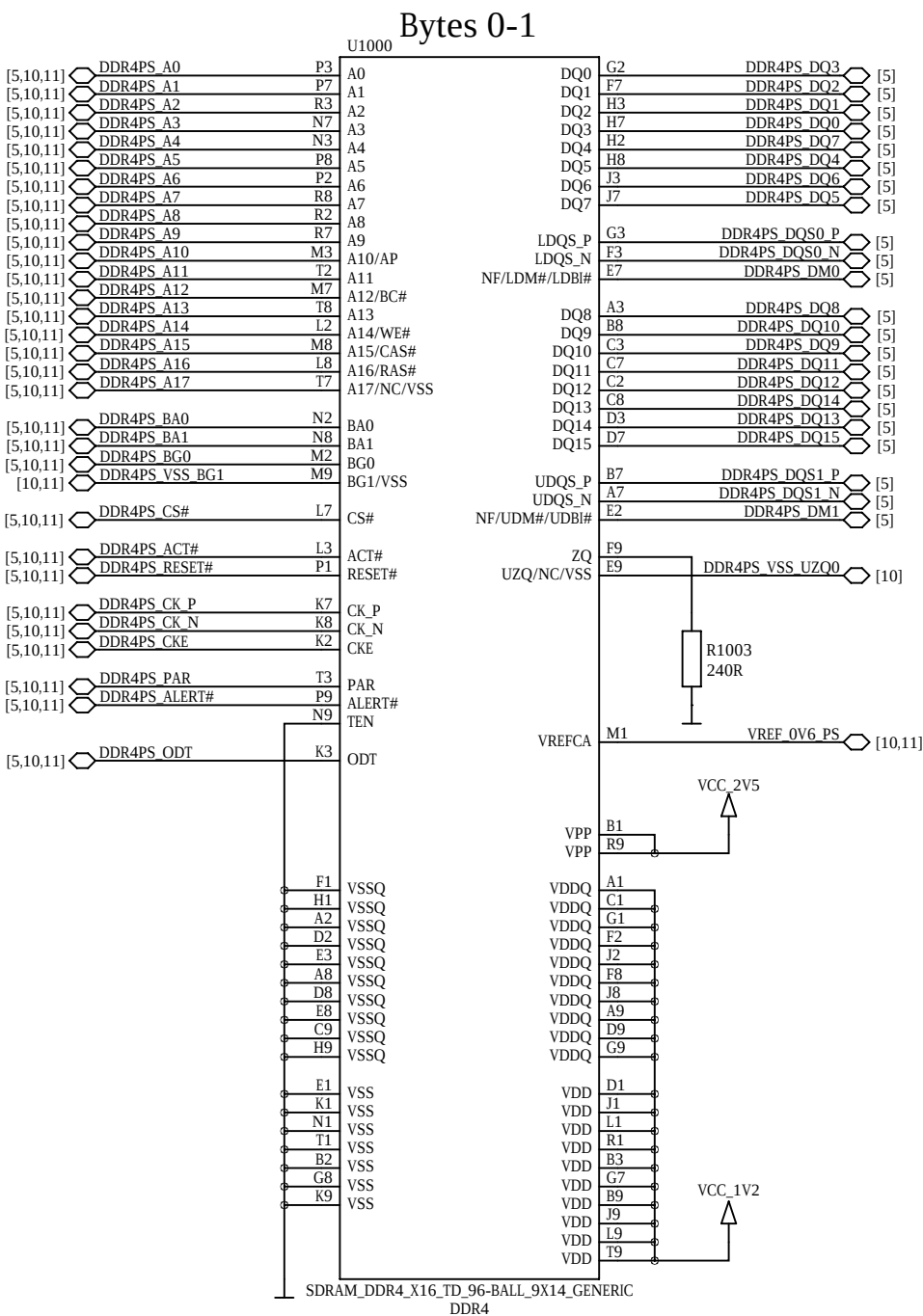
B

B

A

A

PS DDR4 SDRAM

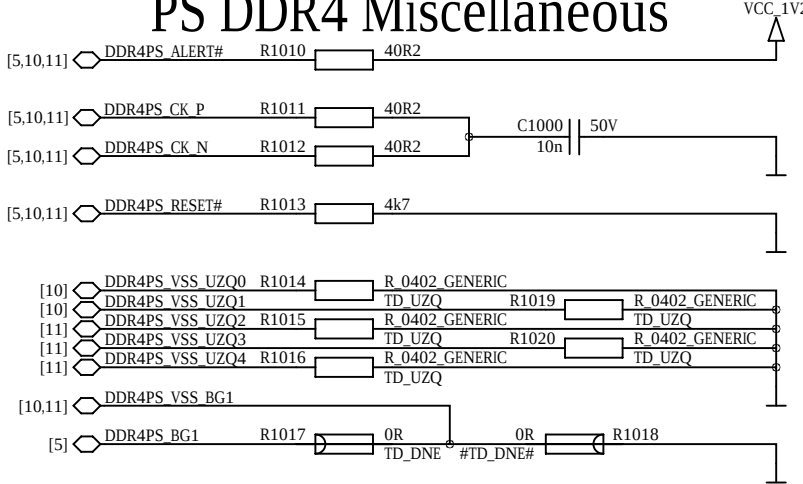


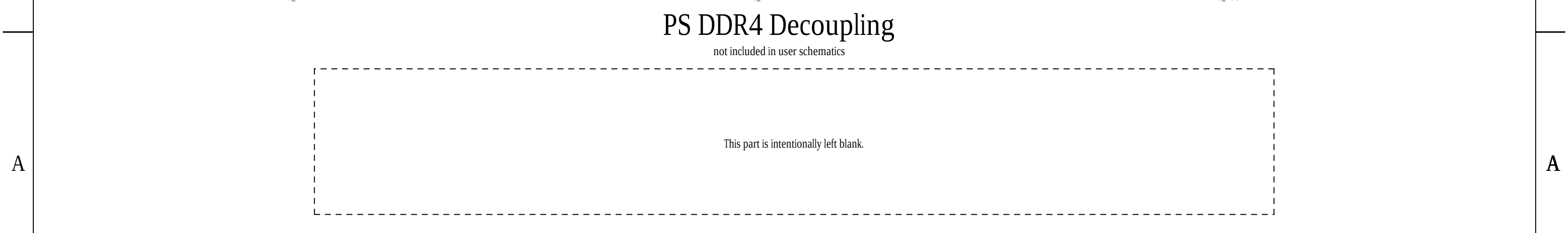
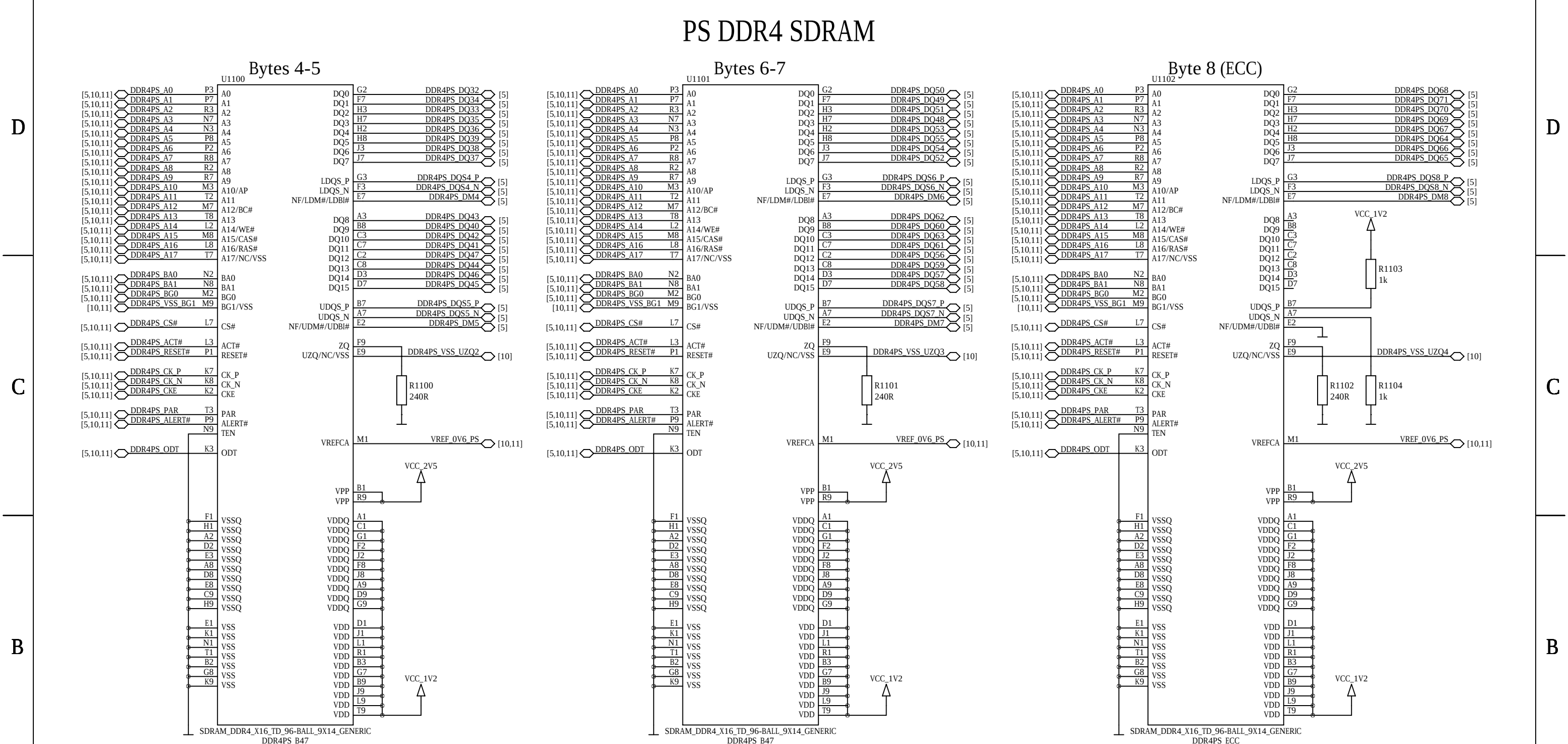
PS DDR4 Termination

not included in user schematics

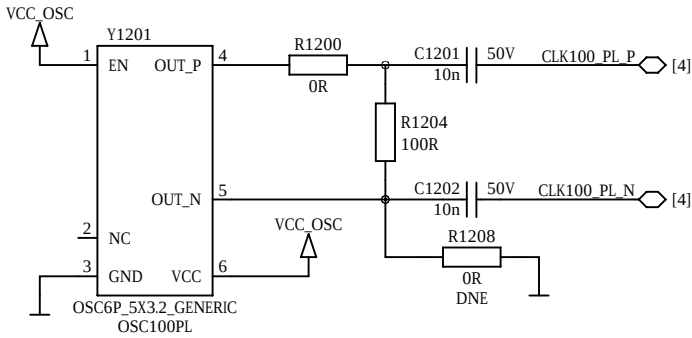
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PS DDR4 Miscellaneous

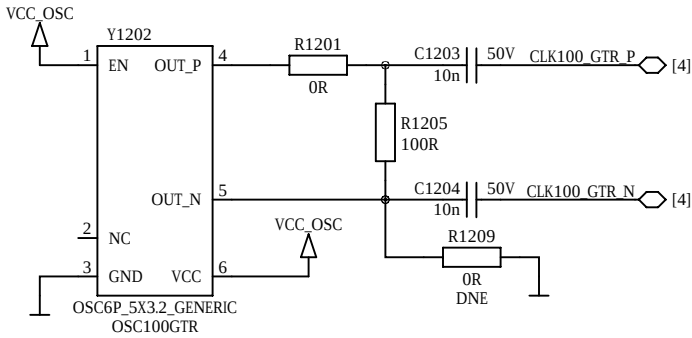




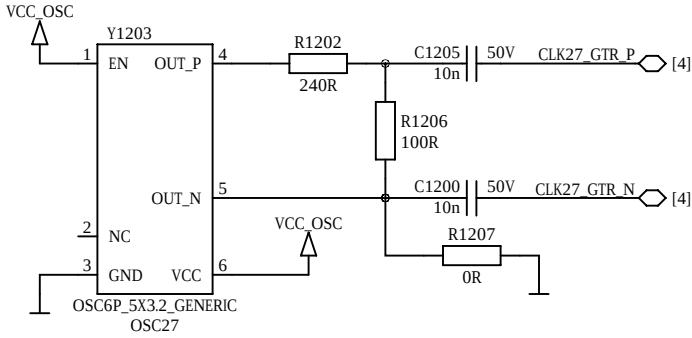
PL 100 MHz Oscillator



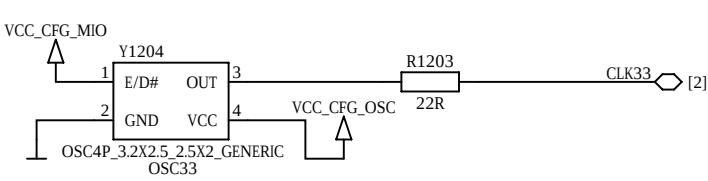
GTR 100 MHz Oscillator



GTR 27 MHz Oscillator



PS 33.3 MHz Oscillator

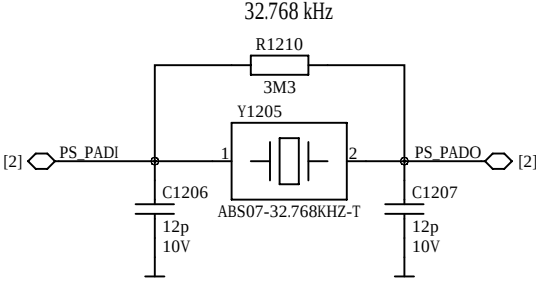


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Oscillator Power Filters

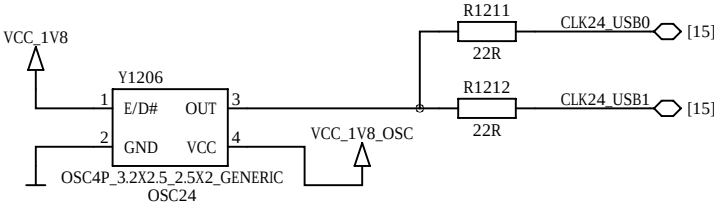
not included in user schematics

RTC Crystal

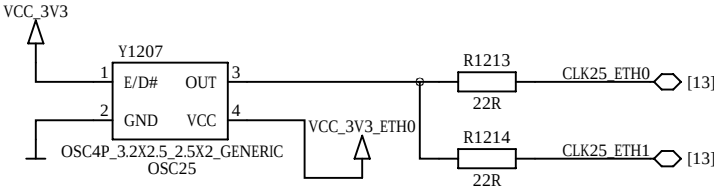


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USB 24 MHz Oscillator



ETH 25 MHz Oscillator



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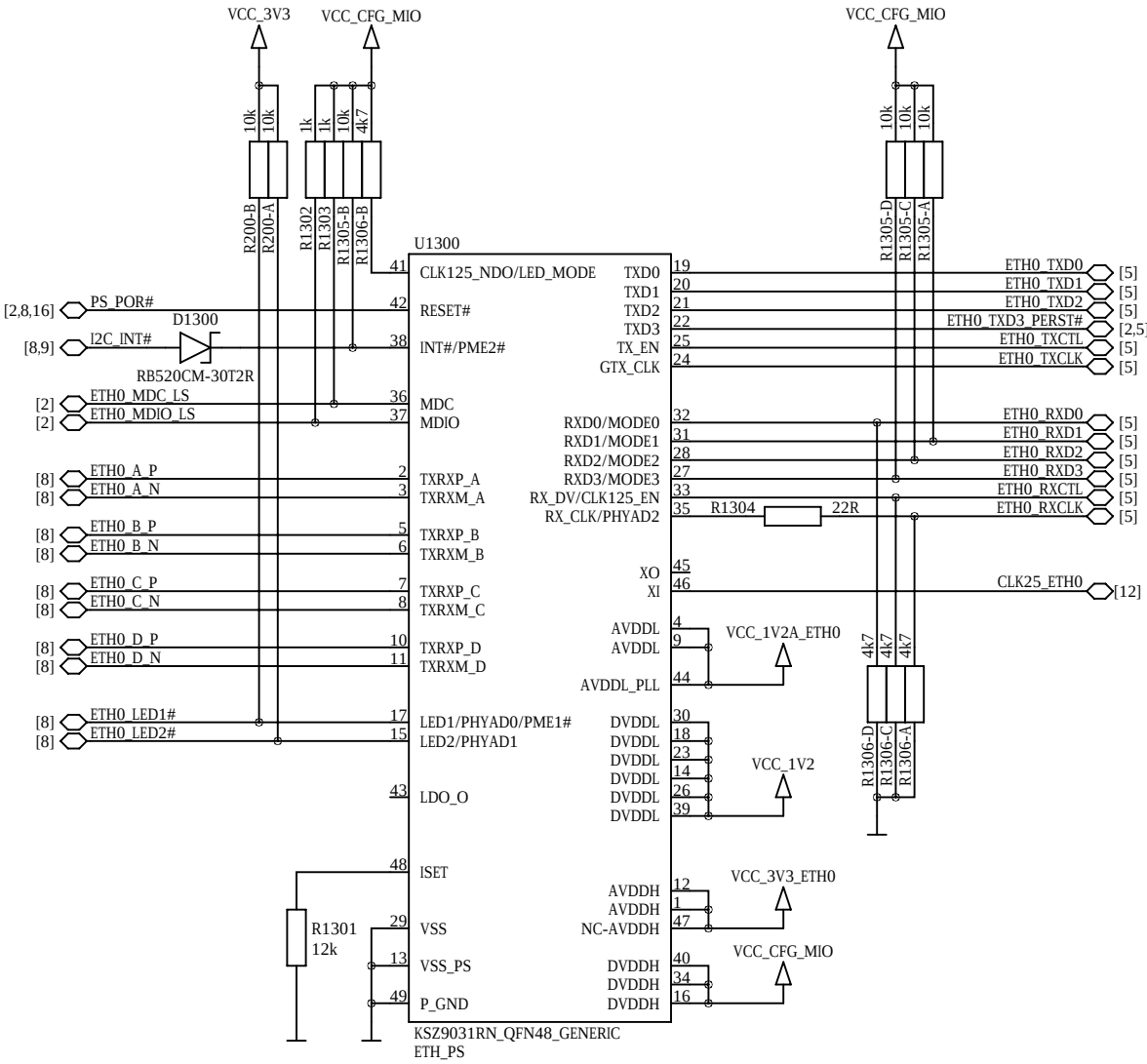
B

A

A

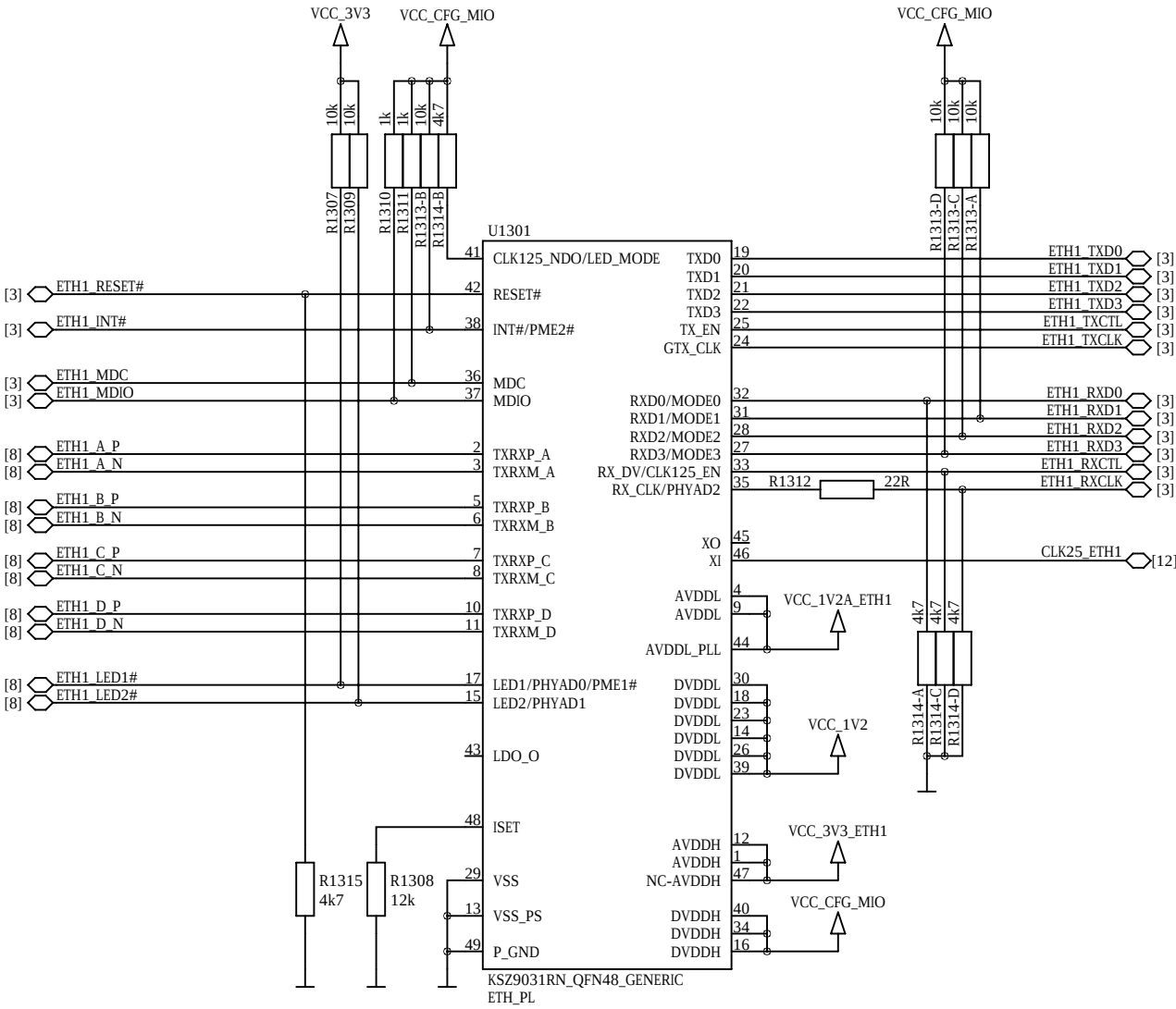
Gigabit Ethernet PHY 0 on PS

not available when using PS PCIe with dedicated PERST# input



MODE<3:0> = 1110 -> RGMII, all capabilities
LED_Mode Pull up = Single LED
PhyAddress <2:0> = 011 -> 3
LED1 and LED2 are active low

Gigabit Ethernet PHY 1 on PL



MODE<3:0> = 1110 -> RGMII, all capabilities
LED_Mode Pull up = Single LED
PhyAddress <2:0> = 011 -> 3
LED1 and LED2 are active low

Ethernet Power Filters

not included in user schematics

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Copyright	© 2021 by Enclustra GmbH	Sheet Name	13_DUAL_GIGABIT_ETHERNET	Customer No	0000	Revision	R1.2	DNE = do not equip	Confidential
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IO / MGT Selection

D

C

B

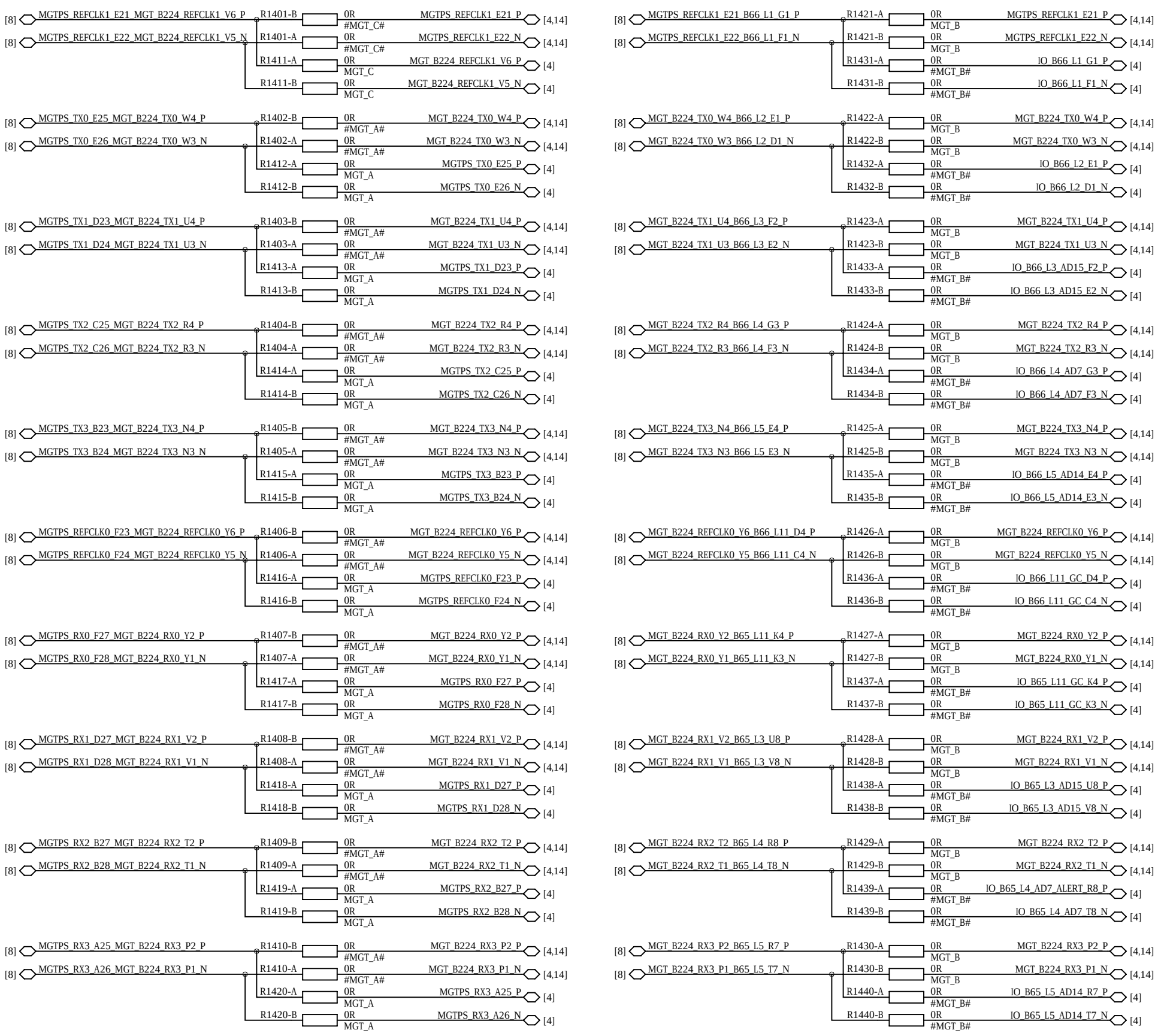
A

D

C

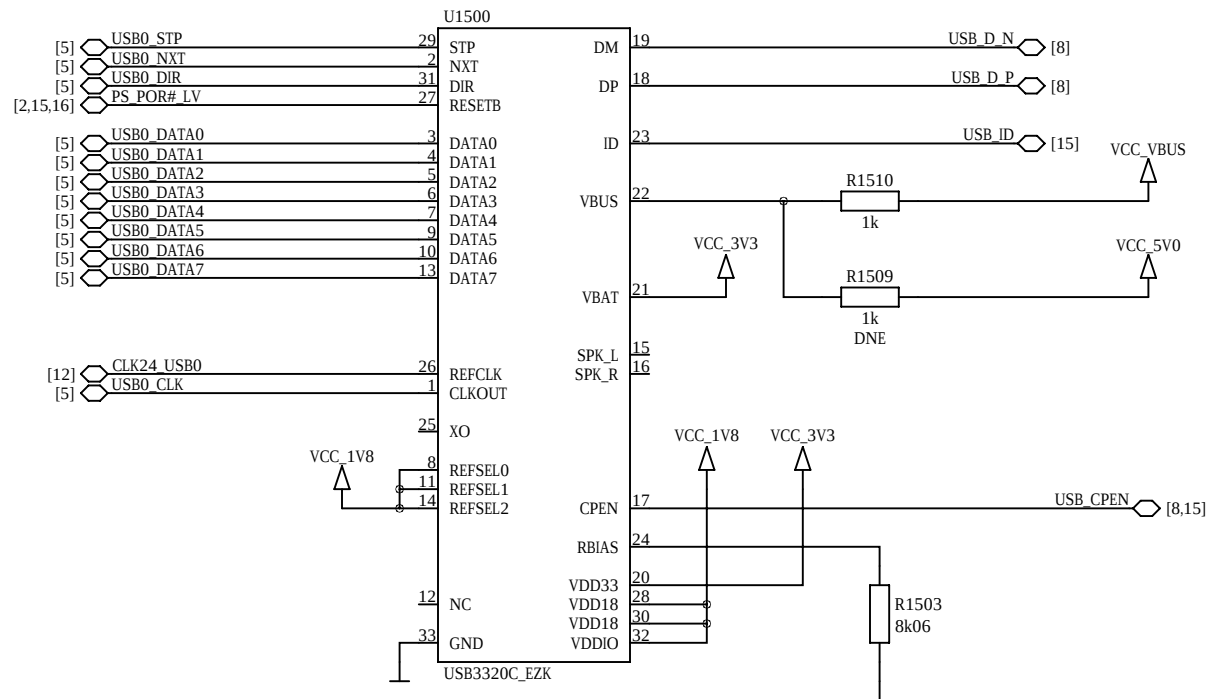
B

A

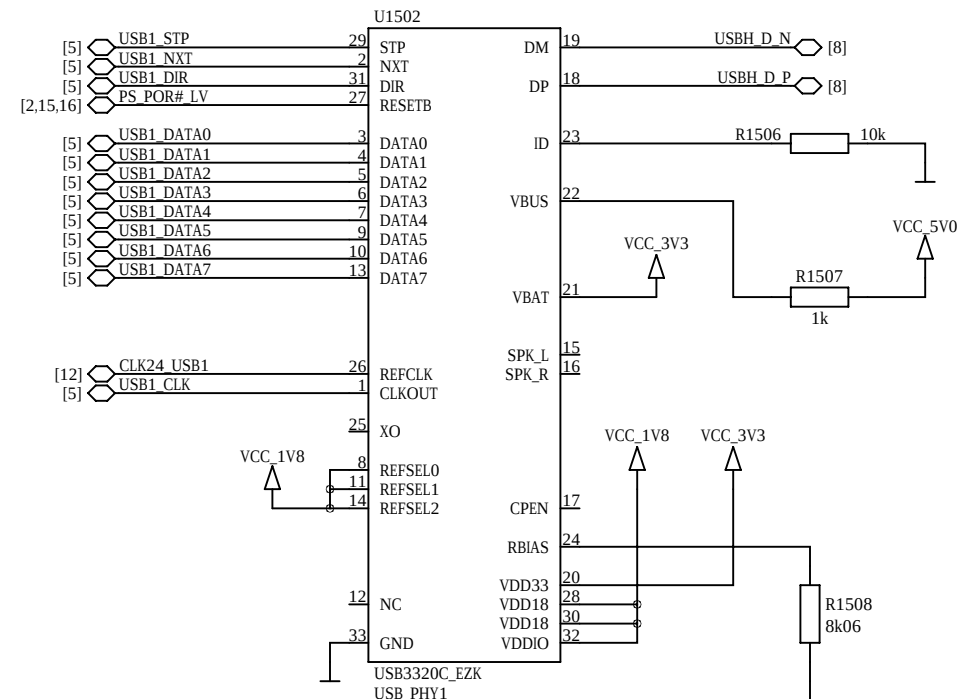


USB 2.0 Device/Host PHY 0

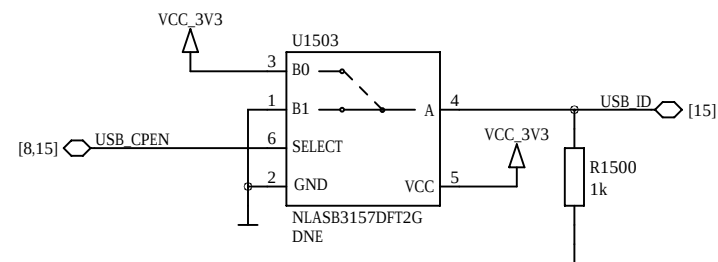
A-Device ID=GND, B-Device ID=Float, Not Used ID=VDD33



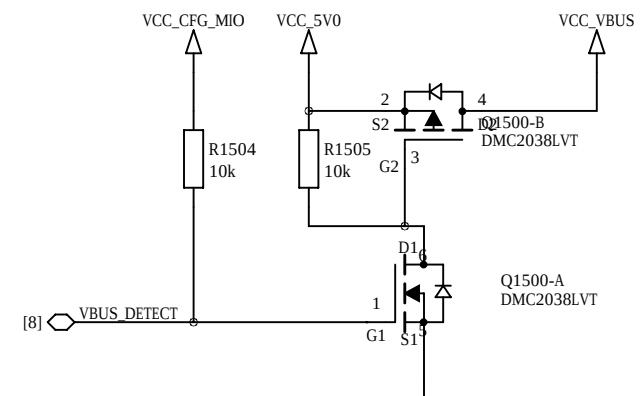
USB 2.0 Host PHY 1



USB 2.0 Host/Device Mode



USB 2.0 VBUS Detect



USB 2.0 Power Decoupling

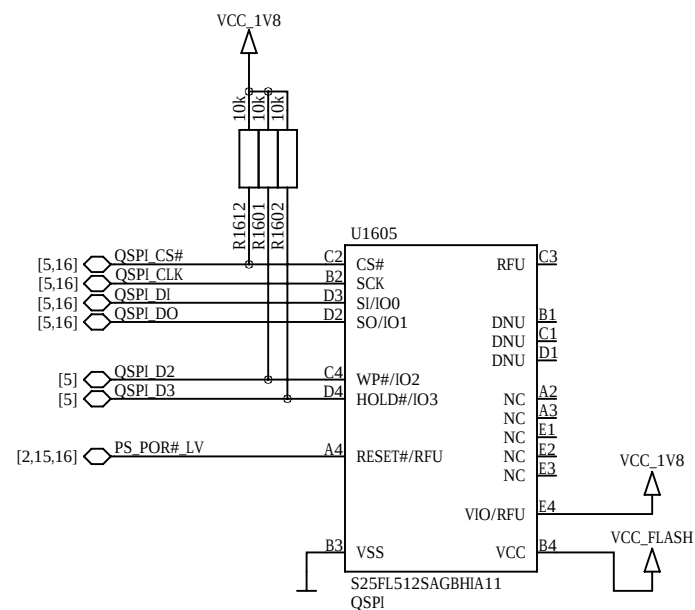
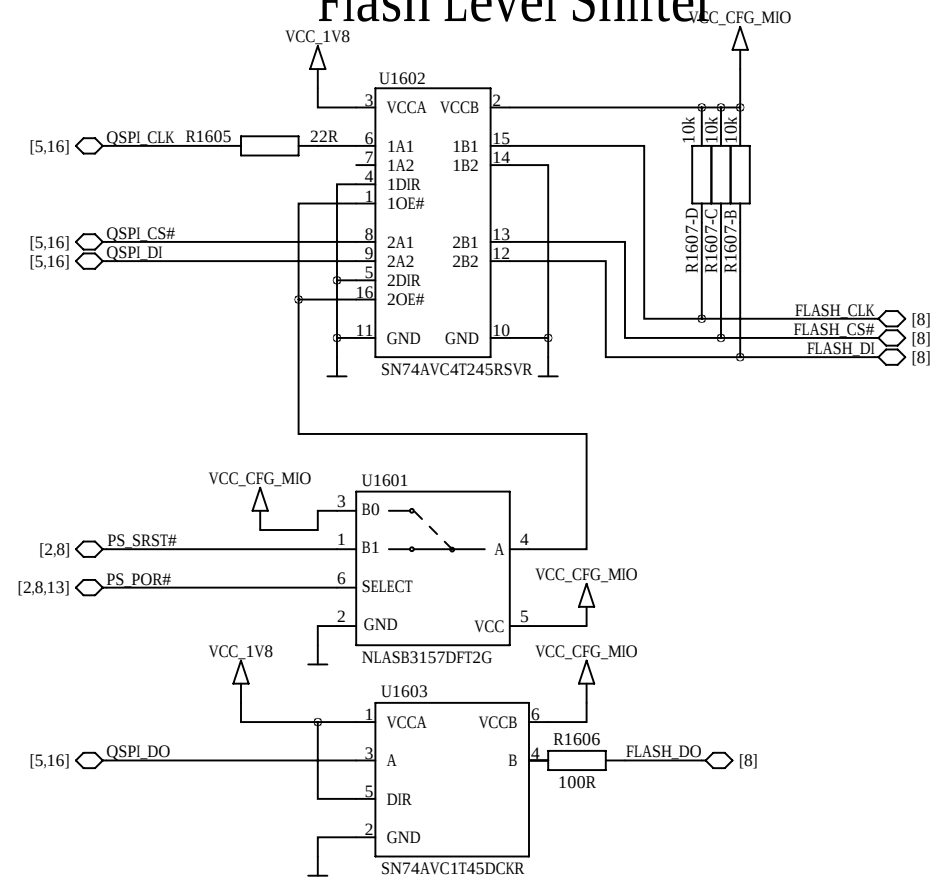
not included in user schematics

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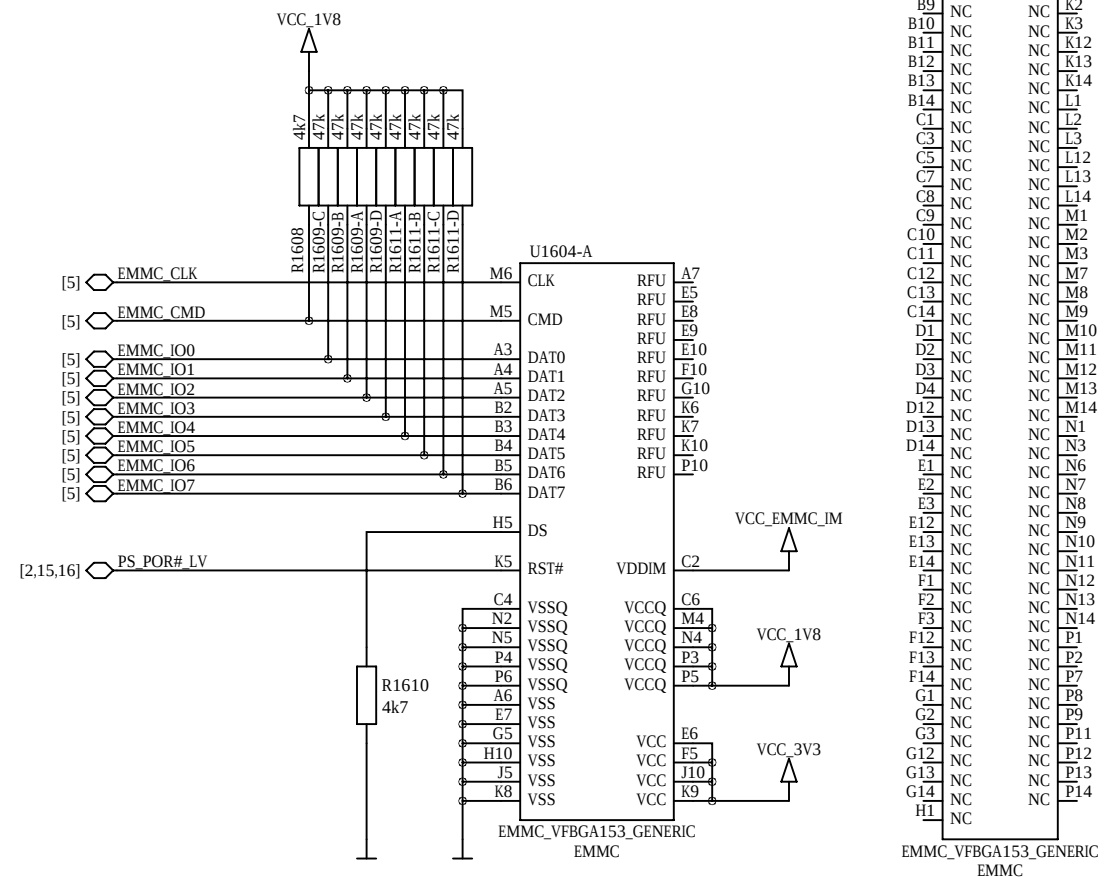
Quad SPI Flash

pin compatible with Micron N25Q512A11G1240F

Micron requires VCC_FLASH = 1.8V

Flash Level Shifter_{VCC}

eMMC Flash



Quad SPI Power

not included in user schematics

This part is intentionally left blank.

eMMC Power

not included in user schematics

This part is intentionally left blank.



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3

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1

D

C

B

A

DC / DC Converter 3.3V 9A 5%

not included in user schematics

This part is intentionally left blank.

Supported Voltage Ranges

Power Rail	Direction	Range	Tolerance
VCC_MOD	input	5 - 15V	±5%
VCC_CFG_MIO	input	1.8 - 3.3V	±5%
VCC_IO_BN	input	1.2 - 3.3V	±5%
VCC_IO_BO	input	1.2 - 3.3V	±5%
VCC_IO_B65	input	1.0 - 1.8V	±5%
VCC_IO_B66	input	1.0 - 1.8V	±5%
VCC_BAT	input	2.7 - 3.6V *	-
VCC_1V8	output	1.8V	±5%
VCC_2V5	output	2.5V	±5%
VCC_3V3	output	3.3V	±5%

When using VCC_IO_BN, VCC_IO_BO at 3.3V, the tolerance is -5% +3%.

* Voltage range valid for LDO assembly option for VCC_BAT_FPGA (default assembly variant).

DC / DC Converter 0.72/0.85/0.9V 9A 3%

not included in user schematics

This part is intentionally left blank.

Input Power Filter

not included in user schematics

This part is intentionally left blank.

LDO 5.0V 0.15A

not included in user schematics

This part is intentionally left blank.



Copyright	© 2021 by Enclustra GmbH	Sheet Name	17_POWER_INT_3.3V_5.0V	Customer No	0000	Revision	R1.2	DNE = do not equip	Confidential		
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Buck Converter 1.2V 3A 3%

not included in user schematics

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DDR4 Termination Voltage Regulator

not included in user schematics

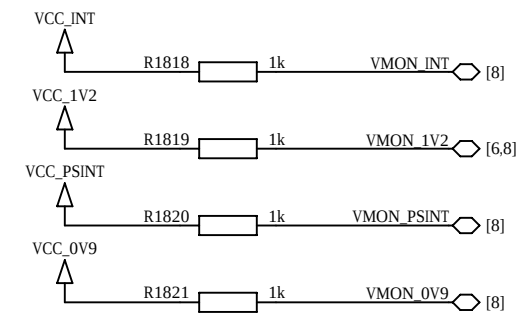
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Buck Converter 0.85/0.9V 6A 3%

not included in user schematics

This part is intentionally left blank.

Voltage Monitoring



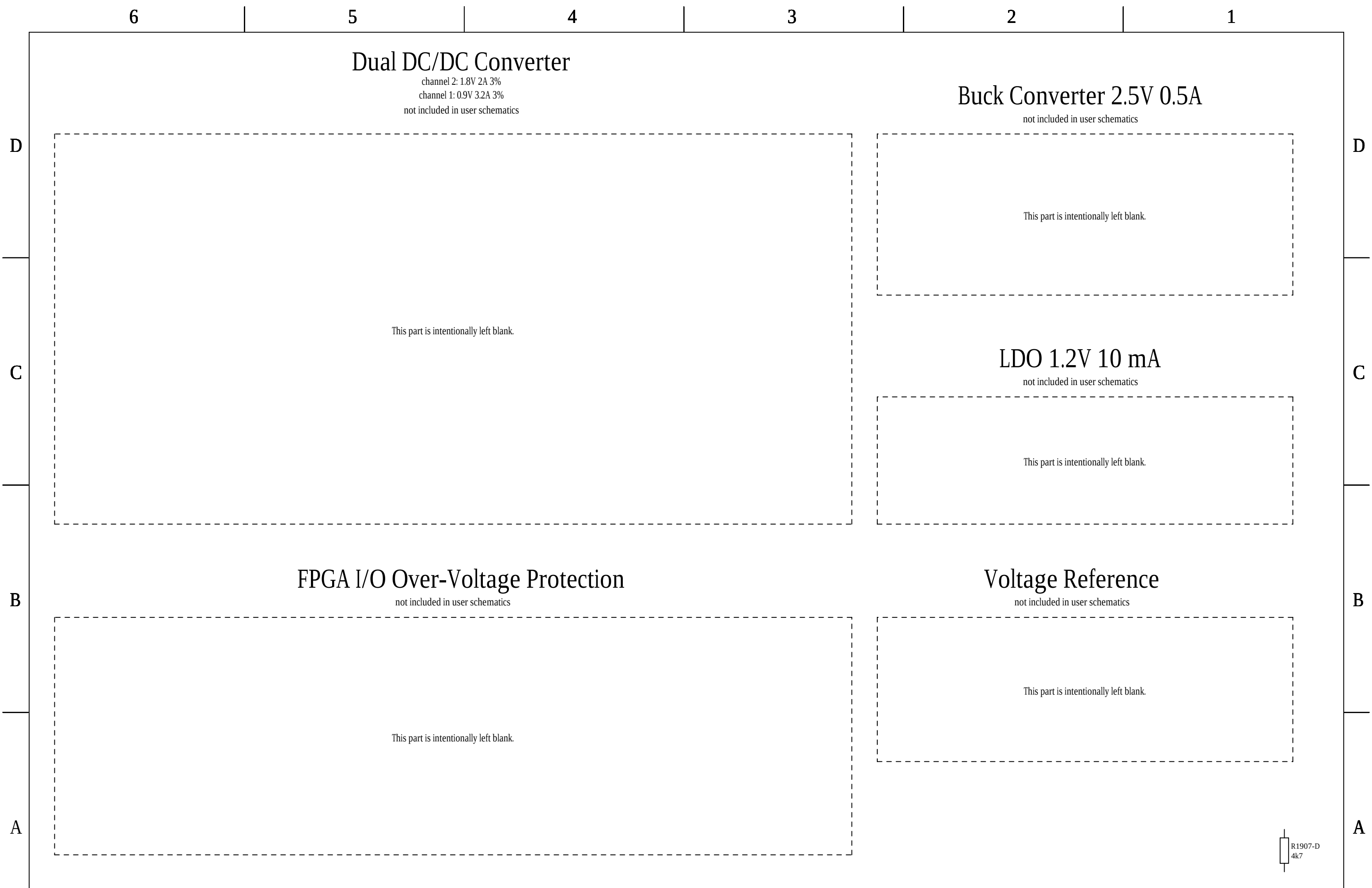
DDR4 VREF Generation

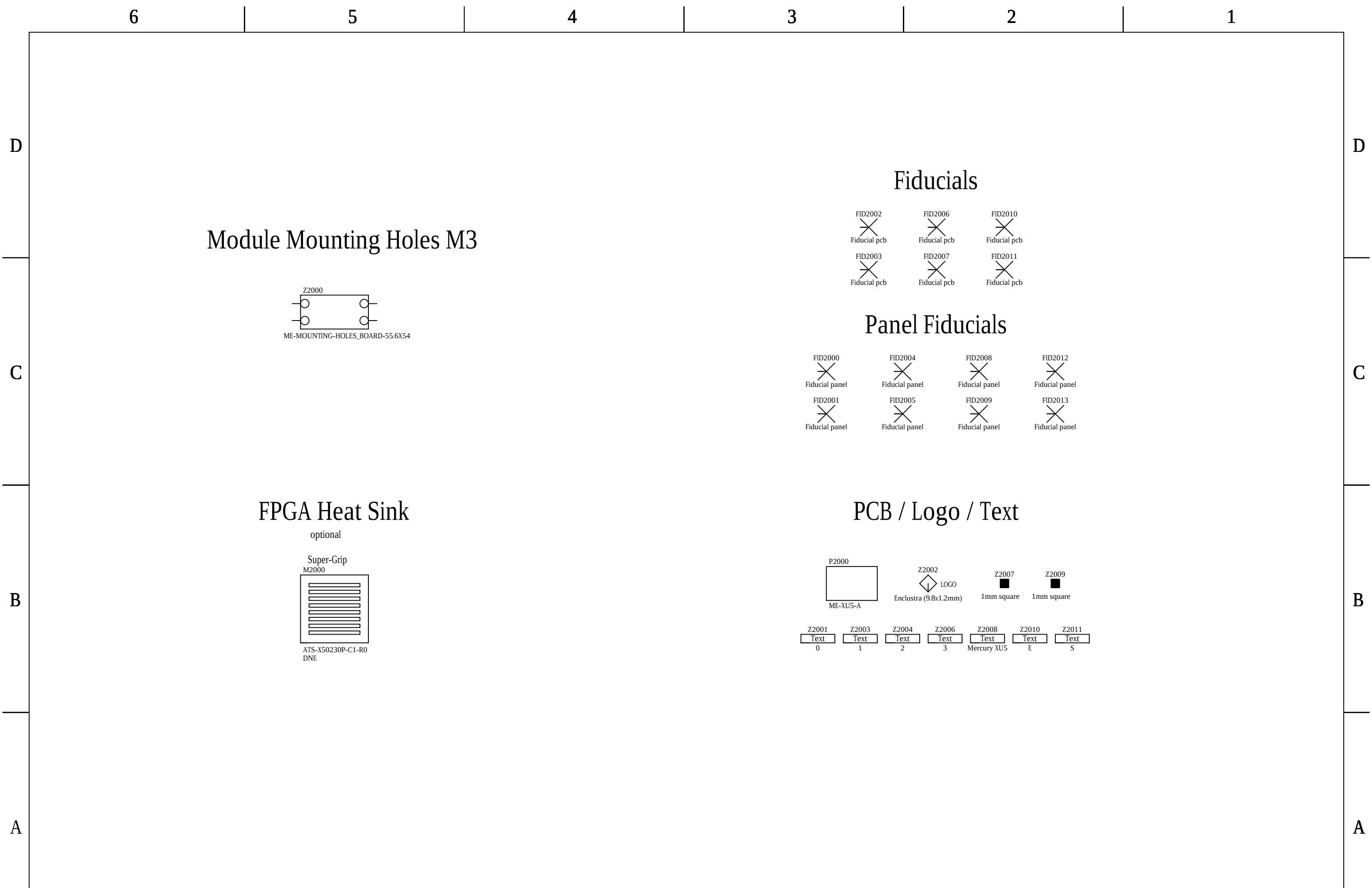
not included in user schematics

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Customer No

0000

Project No

445

Revision

R1.2

Designed

MHEI

DNE = do not equip

Date

17 Mai 2021

Confidential

Sheet/sheets

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D

C

Component Assembly

B

A

FPGA	DDR4	DDR4PS_B47	DDR4PS_ECC	DDR4_PL	eMMC	QSPI	MGT_A	MGT_B	MGT_C	Ethernet PHY PS	Ethernet PHY PL
Part U200A XCZU2CG-1SFVC784E EN102619 Part U200B XCZU2EG-1SFVC784I EN101958 Part U200C XCZU3EG-2SFVC784I EN102020 Part U200D XCZU4EV-1SFVC784I EN102021 Part U200E XCZU5EV-2SFVC784I EN101987 Part U200F XCZU2EG-1SFVC784I4524 EN104000	Part U1000A NDQ46PFP-8NIT EN103227 U1000B MT40A256M16GE-083E EN101632 U1000C AS4C512M16D4-75BIN EN104035 U1000D MT40A1G16KNR-075E EN102109 U1000E K4AAG165WB-MCRC EN102970 Part U1001A NDQ46PFP-8NIT EN103227 U1001B MT40A256M16GE-083E EN101632 U1001C AS4C512M16D4-75BIN EN104035 U1001D MT40A1G16KNR-075E EN102109 U1001E K4AAG165WB-MCRC EN102970	Part U1100A NDQ46PFP-8NIT EN103227 U1100B MT40A256M16GE-083E EN101632 U1100C AS4C512M16D4-75BIN EN104035 U1100D MT40A1G16KNR-075E EN102109 U1100E K4AAG165WB-MCRC EN102970 Part U1101A NDQ46PFP-8NIT EN103227 U1101B MT40A256M16GE-083E EN101632 U1101C AS4C512M16D4-75BIN EN104035 U1101D MT40A1G16KNR-075E EN102109 U1101E K4AAG165WB-MCRC EN102970	Part U1102A NDQ46PFP-8NIT EN103227 U1102B MT40A256M16GE-083E EN101632 U1102C AS4C512M16D4-75BIN EN104035 U1102D MT40A1G16KNR-075E EN102109 U1102E K4AAG165WB-MCRC EN102970 Part U1102A NDQ46PFP-8NIT EN103227 U1102B MT40A256M16GE-083E EN101632 U1102C AS4C512M16D4-75BIN EN104035 U1102D MT40A1G16KNR-075E EN102109 U1102E K4AAG165WB-MCRC EN102970	Part U900A NDQ46PFP-8NIT EN103227 U900B MT40A256M16GE-083E EN101632 U900C AS4C512M16D4-75BIN EN104035 U900D MT40A1G16KNR-075E EN102109 U900E K4AAG165WB-MCRC EN102970 Part U900A NDQ46PFP-8NIT EN103227 U900B MT40A256M16GE-083E EN101632 U900C AS4C512M16D4-75BIN EN104035 U900D MT40A1G16KNR-075E EN102109 U900E K4AAG165WB-MCRC EN102970	Part U1604A H26M52208FPRI EN101865 Part U1604B EMMC16G-IB29-PZ90 EN102591 Part U1604C IMEM064GT25BM2D EN104001		Part R1402 - R1410 or R1412 - R1420	Part R218, R220, R1421 - R1430 or R217, R219, R1431 - R1440	Part R1401 or R1411	Part U1300A KSZ9031RNXIA EN101934	Part U1301A KSZ9031RNXIA EN101934

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Component Options

SEC_EEPROM	OSC25	OSC100PL	OSC100GTR	OSC27	OSC33	OSC24	L0U68	L1U	L1U_2520	TD_UZQ	TD_DNE
Part U902A ATSHA204A-MAHDA-T EN100827 Part U902B DS28CNO1U-A00+ EN100280	Part Y1207A ASDMB-25.000MHZ-LC-T EN101588 Part Y1207B ASEMB_25.000MHZ-LC-T EN101589 Part Y1207C ASE-25.000MHZ-LC-T EN100287 Part Y1207D 831055678 EN104082	Part Y1201A ASDMPLV-100.000MHZ-LR-T3 EN101610 Part Y1201B ASFLMPLV-100.000MHZ-LR-T EN101611	Part Y1202A ASDMPLV-100.000MHZ-LR-T3 EN101610 Part Y1202B ASFLMPLV-100.000MHZ-LR-T EN101611	Part Y1203A ASEMPC_27.000MHZ-LR-T EN101608 Part Y1203B ASEMPC-27.000MHZ-XR-T EN101609	Part Y1204A ASEMB_33.333MHZ-LC-T EN100914 Part Y1204B ASDMB_33.333MHZ-LC-T EN100285	Part Y1206A ASDMB-24.000MHZ-LC-T EN101221 Part Y1206B ASEMB_24.000MHZ-LC-T EN101187 Part Y1206C O_24.0-JO32-B-1V3-1-T1-LF EN104079 Part Y1206D ECS-3225MVQ-240-BP-TR EN104080	Part L1700A 0u68 EN100848 Part L1700B 1u EN100436 Part L1701A 0u68 EN100848 Part L1701B 1u EN100436	Part L1801A 1u EN104020 Part L1801B 0u68 EN102445	Part L1903A 1u0 EN101851 Part L1903B 1u0 EN101811	Part R906A 0R EN101361 Part R906B 240R EN101382 Part R1014A 0R EN101361 Part R1014B 240R EN101382 Part R1015A 0R EN101361 Part R1015B 240R EN101382 Part R1016A 0R EN101361 Part R1016B 240R EN101382 Part R1019A 0R EN101361 Part R1019B 240R EN101382 Part R1020A 0R EN101361 Part R1020B 240R EN101382	Option PL DDR: R902 or R903, R907 Option PS DDR: R1004, R1017 or R1018

Component Assembly

SEC_EEPROM	OSC25	OSC100PL	OSC100GTR	OSC27	OSC33	OSC24	L0U68	L1U	L1U_2520	TD_UZQ	TD_DNE
Option EN102596:8 SEC_EEPROM EN100827 Option EN102224:8 SEC_EEPROM EN100827 Option EN102225:8 SEC_EEPROM EN100827 Option EN102226:8 SEC_EEPROM EN100827 Option EN102285:8 SEC_EEPROM EN100827 Option EN102227:8 SEC_EEPROM EN100827 Option EN102229:8 SEC_EEPROM EN100827 Option EN103062:8 SEC_EEPROM EN100827 Option EN103063:8 SEC_EEPROM EN100827 Option EN103077:8 SEC_EEPROM EN100827 Option EN103308:8 SEC_EEPROM EN100827 Option EN104005:8 SEC_EEPROM EN100827 Option EN104016:8 SEC_EEPROM EN100827	Option EN102596:9 OSC25 EN100287 Option EN102224:9 OSC25 EN100287 Option EN102225:9 OSC25 EN100287 Option EN102226:9 OSC25 EN100287 Option EN102285:9 OSC25 EN100287 Option EN102227:9 OSC25 EN100287 Option EN102229:9 OSC25 EN100287 Option EN103062:9 OSC25 EN100287 Option EN103063:9 OSC25 EN100287 Option EN103077:9 OSC25 EN100287 Option EN103308:9 OSC25 EN100287 Option EN104005:9 OSC25 EN100287 Option EN104016:9 OSC25 EN100287	Option EN102596:10 OSC100PL EN101611 Option EN102224:10 OSC100PL EN101611 Option EN102225:10 OSC100PL EN101611 Option EN102226:10 OSC100PL EN101611 Option EN102285:10 OSC100PL EN101611 Option EN102227:10 OSC100PL EN101611 Option EN102229:10 OSC100PL EN101611 Option EN103062:10 OSC100PL EN101611 Option EN103063:10 OSC100PL EN101611 Option EN103077:10 OSC100PL EN101611 Option EN103308:10 OSC100PL EN101611 Option EN104005:10 OSC100PL EN101611 Option EN104016:10 OSC100PL EN101611	Option EN102596:11 OSC100GTR EN101611 Option EN102224:11 OSC100GTR EN101611 Option EN102225:11 OSC100GTR EN101611 Option EN102226:11 OSC100GTR EN101611 Option EN102285:11 OSC100GTR EN101611 Option EN102227:11 OSC100GTR EN101611 Option EN102229:11 OSC100GTR EN101611 Option EN103062:11 OSC100GTR EN101611 Option EN103063:11 OSC100GTR EN101611 Option EN103077:11 OSC100GTR EN101611 Option EN103308:11 OSC100GTR EN101611 Option EN104005:11 OSC100GTR EN101611 Option EN104016:11 OSC100GTR EN101611	Option EN102596:12 OSC27 EN101608 Option EN102224:12 OSC27 EN101608 Option EN102225:12 OSC27 EN101608 Option EN102226:12 OSC27 EN101608 Option EN102285:12 OSC27 EN101608 Option EN102227:12 OSC27 EN101608 Option EN102229:12 OSC27 EN101608 Option EN103062:12 OSC27 EN101608 Option EN103063:12 OSC27 EN101608 Option EN103077:12 OSC27 EN101608 Option EN103308:12 OSC27 EN101608 Option EN104005:12 OSC27 EN101608 Option EN104016:12 OSC27 EN101608	Option EN102596:13 OSC33 EN100914 Option EN102224:13 OSC33 EN100914 Option EN102225:13 OSC33 EN100914 Option EN102226:13 OSC33 EN100914 Option EN102285:13 OSC33 EN100914 Option EN102227:13 OSC33 EN100914 Option EN102229:13 OSC33 EN100914 Option EN103062:13 OSC33 EN100914 Option EN103063:13 OSC33 EN100914 Option EN103077:13 OSC33 EN100914 Option EN103308:13 OSC33 EN100914 Option EN104005:13 OSC33 EN100914 Option EN104016:13 OSC33 EN100914	Option EN102596:14 OSC24 EN104079 Option EN102224:14 OSC24 EN104079 Option EN102225:14 OSC24 EN104079 Option EN102226:14 OSC24 EN104079 Option EN102285:14 OSC24 EN104079 Option EN102227:14 OSC24 EN104079 Option EN102229:14 OSC24 EN104079 Option EN103062:14 OSC24 EN104079 Option EN103063:14 OSC24 EN104079 Option EN103077:14 OSC24 EN104079 Option EN103308:14 OSC24 EN104079 Option EN104005:14 OSC24 EN104079 Option EN104016:14 OSC24 EN104079	Option EN102596:15 L0U68 EN100848 Option EN102224:15 L0U68 EN100848 Option EN102225:15 L0U68 EN100848 Option EN102226:15 L0U68 EN100848 Option EN102285:15 L0U68 EN100848 Option EN102227:15 L0U68 EN100848 Option EN102229:15 L0U68 EN100848 Option EN103062:15 L0U68 EN100848 Option EN103063:15 L0U68 EN100848 Option EN103077:15 L0U68 EN100848 Option EN103308:15 L0U68 EN100848 Option EN104005:15 L0U68 EN100848 Option EN104016:15 L0U68 EN100848	Option EN102596:16 L1U EN104020 Option EN102224:16 L1U EN104020 Option EN102225:16 L1U EN104020 Option EN102226:16 L1U EN104020 Option EN102285:16 L1U EN104020 Option EN102227:16 L1U EN104020 Option EN102229:16 L1U EN104020 Option EN103062:16 L1U EN104020 Option EN103063:16 L1U EN104020 Option EN103077:16 L1U EN104020 Option EN103308:16 L1U EN104020 Option EN104005:16 L1U EN104020 Option EN104016:16 L1U EN104020	Option EN102596:17 L1U_2520 EN101811 Option EN102224:17 L1U_2520 EN101811 Option EN102225:17 L1U_2520 EN101811 Option EN102226:17 L1U_2520 EN101811 Option EN102285:17 L1U_2520 EN101811 Option EN102227:17 L1U_2520 EN101811 Option EN102229:17 L1U_2520 EN101811 Option EN103062:17 L1U_2520 EN101811 Option EN103063:17 L1U_2520 EN101811 Option EN103077:17 L1U_2520 EN101811 Option EN103308:17 L1U_2520 EN101811 Option EN104005:17 L1U_2520 EN101811 Option EN104016:17 L1U_2520 EN101811	Option EN102596:18 TD_UZQ EN101361 Option EN102224:18 TD_UZQ EN101361 Option EN102225:18 TD_UZQ EN101361 Option EN102226:18 TD_UZQ EN101361 Option EN102285:18 TD_UZQ EN101361 Option EN102227:18 TD_UZQ EN101361 Option EN102229:18 TD_UZQ EN101361 Option EN103062:18 TD_UZQ EN101361 Option EN103063:18 TD_UZQ EN101382 Option EN103077:18 TD_UZQ EN101361 Option EN103308:18 TD_UZQ EN101361 Option EN104005:18 TD_UZQ EN101361 Option EN104016:18 TD_UZQ EN101361	Option EN102596:19 TD_DNE DNE Option EN102224:19 TD_DNE DNE Option EN102225:19 TD_DNE DNE Option EN102226:19 TD_DNE DNE Option EN102285:19 TD_DNE DNE Option EN102227:19 TD_DNE DNE Option EN102229:19 TD_DNE DNE Option EN103062:19 TD_DNE DNE Option EN103063:19 TD_DNE DNE Option EN103077:19 TD_DNE DNE Option EN103308:19 TD_DNE DNE Option EN104005:19 TD_DNE DNE Option EN104016:19 TD_DNE DNE

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Component Options											
L2U2_2520		LOU68_5050		L1U5_2520		JMOD		USB_PHY1			
Part L1901A 2u2 EN101850		Part L1800A 0u68 EN102167		Part L1902A 1u5 EN102212		Part J800A FX10A-168S-SV EN100375		U1502 or DNE			
Part L1901B 2u2 EN101849		Part L1800B 0u68 EN102168		Part J800B FX10A-168S-SV(83) EN103313							
				Part J801A FX10A-168S-SV EN100375							
				Part J801B FX10A-168S-SV(83) EN103313							
Component Assembly											
L2U2_2520		LOU68_5050		L1U5_2520		JMOD		USB_PHY1			
Option EN102596:20 L2U2_2520 EN101850		Option EN102596:21 LOU68_5050 EN102168		Option EN102596:22 L1U5_2520 EN102212		Option EN102596:28 JMOD EN100375		Option EN102596:29 USB_PHY1			
Option EN102224:20 L2U2_2520 EN101850		Option EN102224:21 LOU68_5050 EN102168		Option EN102224:22 L1U5_2520 EN102212		Option EN102224:28 JMOD EN100375		Option EN102224:29 USB_PHY1			
Option EN102225:20 L2U2_2520 EN101850		Option EN102225:21 LOU68_5050 EN102168		Option EN102225:22 L1U5_2520 EN102212		Option EN102225:28 JMOD EN100375		Option EN102225:29 USB_PHY1			
Option EN102226:20 L2U2_2520 EN101850		Option EN102226:21 LOU68_5050 EN102168		Option EN102226:22 L1U5_2520 EN102212		Option EN102226:28 JMOD EN100375		Option EN102226:29 USB_PHY1			
Option EN102285:20 L2U2_2520 EN101850		Option EN102285:21 LOU68_5050 EN102168		Option EN102285:22 L1U5_2520 EN102212		Option EN102285:28 JMOD EN100375		Option EN102285:29 USB_PHY1			
Option EN102227:20 L2U2_2520 EN101850		Option EN102227:21 LOU68_5050 EN102168		Option EN102227:22 L1U5_2520 EN102212		Option EN102227:28 JMOD EN100375		Option EN102227:29 USB_PHY1			
Option EN102229:20 L2U2_2520 EN101850		Option EN102229:21 LOU68_5050 EN102168		Option EN102229:22 L1U5_2520 EN102212		Option EN102229:28 JMOD EN100375		Option EN102229:29 USB_PHY1			
Option EN103062:20 L2U2_2520 EN101850		Option EN103062:21 LOU68_5050 EN102168		Option EN103062:22 L1U5_2520 EN102212		Option EN103062:28 JMOD EN100375		Option EN103062:29 USB_PHY1			
Option EN103063:20 L2U2_2520 EN101850		Option EN103063:21 LOU68_5050 EN102168		Option EN103063:22 L1U5_2520 EN102212		Option EN103063:28 JMOD EN100375		Option EN103063:29 USB_PHY1			
Option EN103077:20 L2U2_2520 EN101850		Option EN103077:21 LOU68_5050 EN102168		Option EN103077:22 L1U5_2520 EN102212		Option EN103077:28 JMOD EN100375		Option EN103077:29 USB_PHY1			
Option EN103308:20 L2U2_2520 EN101850		Option EN103308:21 LOU68_5050 EN102168		Option EN103308:22 L1U5_2520 EN102212		Option EN103308:28 JMOD EN103313		Option EN103308:29 USB_PHY1			
Option EN104005:20 L2U2_2520 EN101850		Option EN104005:21 LOU68_5050 EN102168		Option EN104005:22 L1U5_2520 EN102212		Option EN104005:28 JMOD EN100375		Option EN104005:29 USB_PHY1			
Option EN104016:20 L2U2_2520 EN101850		Option EN104016:21 LOU68_5050 EN102168		Option EN104016:22 L1U5_2520 EN102212		Option EN104016:28 JMOD EN100375		Option EN104016:29 USB_PHY1 DNE			
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